

NEET(UG) – 2022 (Overseas)

Section-A (Physics)

- The de Broglie wavelength of a thermal electron at 27°C is λ . When the temperature is increased to 927°C , its de-Broglie wavelength will become:
 - 4λ
 - $\frac{\lambda}{2}$
 - $\frac{\lambda}{4}$
 - 2λ
- The ratio of the radii of two circular coils is $1 : 2$. The ratio of currents in the respective coils such that the same magnetic moment is produced at the centre of each coil:
 - $2 : 1$
 - $1 : 2$
 - $1 : 4$
 - $4 : 1$
- The current in an inductor of self inductance 4 H changes from 4 A to 2 A in 1 second . The e.m.f. induced in the coil is:
 - 2 V
 - -4 V
 - 8 V
 - -2 V
- An electromagnetic wave is moving along negative $z(-z)$ direction and at any instant of time, at a point, its electric field vector is $3\hat{j}\text{ V/m}$. The corresponding magnetic field at that point and instant will be:

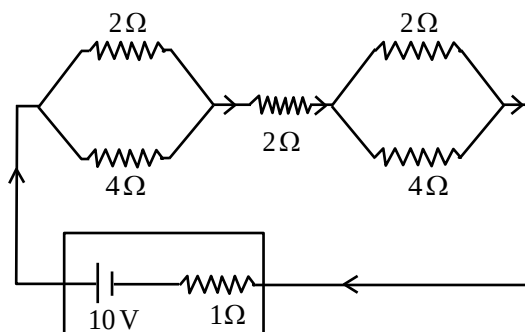
(Take $c = 3 \times 10^8\text{ ms}^{-1}$)

 - $-10\hat{i}\text{ nT}$
 - $\hat{i}\text{ nT}$
 - $-\hat{i}\text{ nT}$
 - $10\hat{i}\text{ nT}$
- Two amplifiers of voltage gain 20 each, are cascaded in series. If 0.01 volt a.c. input signal is applied across the first amplifier, the output a.c. signal of the second amplifier in volt is:
 - 4.0
 - 0.01
 - 0.20
 - 2.0
- A linearly polarised monochromatic light of intensity 10 lumen is incident on a polarizer. The angle between the direction of polarisation of the light and that of the polariser such that the intensity of output light is 2.5 lumen is:
 - 75°
 - 30°
 - 45°
 - 60°

- A stone is thrown vertically downward with an initial velocity of 40 m/s from the top of a building. If it reaches the ground with velocity 60 m/s , then the height of the building is:

(Take $g = 10\text{ m/s}^2$)

 - 140 m
 - 80 m
 - 100 m
 - 120 m
- A concave lens of focal length, 25 cm is sandwiched between two convex lenses, each of focal length, 40 cm . The power in dioptre of the combined lens would be:
 - 9
 - 1
 - 0.01
 - 55
- A network of resistors is connected across a 10 V battery with internal resistance of $1\ \Omega$ as shown in the circuit diagram. The equivalent resistance of the circuit is:

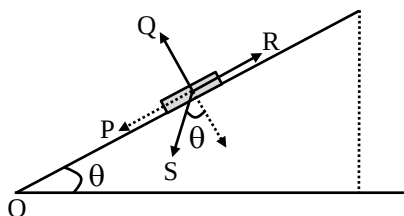


- $\frac{14}{3}\ \Omega$
 - $\frac{12}{7}\ \Omega$
 - $\frac{14}{7}\ \Omega$
 - $\frac{17}{3}\ \Omega$
- The correct statement about the variation of viscosity of fluid with increase in temperature:
 - viscosity of both liquids and gases increases
 - viscosity of liquids increases
 - viscosity of liquids decreases
 - viscosity of gases decreases
 - Let L_1 and L_2 be the orbital angular momentum of an electron in the first and second excited states of hydrogen atom, respectively. According to the Bohr's model, the ratio $L_1 : L_2$ is:
 - $2 : 1$
 - $3 : 2$
 - $2 : 3$
 - $1 : 2$

12. During simple harmonic motion of a body, the energy at the extreme positions is:

(1) is always zero
 (2) purely kinetic
 (3) purely potential
 (4) both kinetic and potential

13.



When a body of mass 'm' just begins to slide as shown, match list-I with List-II:

List-I

(a) Normal reaction
 (b) Frictional force (f_s)
 (c) Weight (mg)
 (d) $mg\sin\theta$

List-II

(i) P
 (ii) Q
 (iii) R
 (iv) S

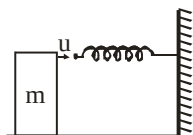
Choose the correct answer from the options given below:

(1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 (2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
 (4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

14. Twelve point charges each of charge q coulomb are placed at the circumference of a circle of radius r with equal angular spacing. If one of the charges is removed, the net electric field (in N/C) at the centre of the circle is : (ϵ_0 – permittivity of free space)

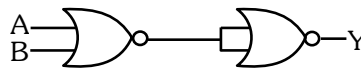
(1) $\frac{13q}{4\pi\epsilon_0 r^2}$ (2) zero
 (3) $\frac{q}{4\pi\epsilon_0 r^2}$ (4) $\frac{12q}{4\pi\epsilon_0 r^2}$

15. A block of mass m is moving with initial velocity u towards a stationary spring of stiffness constant k attached to the wall as shown in the figure. Maximum compression of the spring is:
 (The friction between the block and the surface is negligible)



(1) $4u\sqrt{\frac{m}{k}}$ (2) $2u\sqrt{\frac{m}{k}}$
 (3) $\frac{1}{2}u\sqrt{\frac{k}{m}}$ (4) $u\sqrt{\frac{m}{k}}$

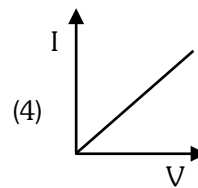
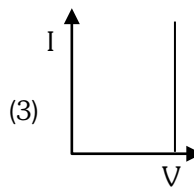
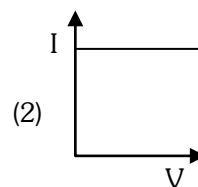
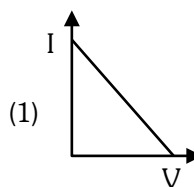
16.



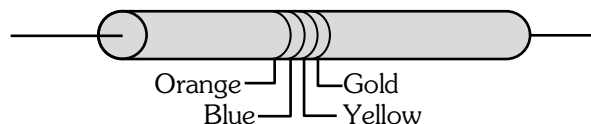
The output of the logic circuit shown is equivalent to a/an:

(1) NOR gate (2) AND gate
 (3) NAND gate (4) OR gate

17. The plot of current I flowing through a conductor versus the applied voltage V across the ends of a conductor is:



18. The value of resistance for the colour code of the given resistor is:



(1) $(470 \pm 47) \text{ k}\Omega$ (2) $(360 \pm 36) \text{ k}\Omega$
 (3) $(360 \pm 18) \text{ k}\Omega$ (4) $(36 \pm 36) \text{ k}\Omega$

19. An inductor coil of self inductance 10 H carries a current of 1 A . The magnetic field energy stored in the coil is:

(1) 2.5 J (2) 20 J
 (3) 5 J (4) 10 J

20. The dimensions of mutual inductance (M) are:

(1) $[MLT^{-2}A^2]$ (2) $[M^2L^2T^{-2}A^2]$
 (3) $[ML^2T^{-2}A^{-2}]$ (4) $[M^2LT^{-2}A^{-2}]$

21. A strong magnetic field is applied along the direction of velocity of an electron. The electron would move along:

(1) the original path (2) a helical path
 (3) a circular path (4) a parabolic path

22. In a photoelectric experiment, blue light is capable of ejecting a photoelectron from a specific metal while green light is not able to eject a photoelectron. Ejection of photoelectrons is also possible using light of the colour:

(1) Red (2) Violet
(3) Orange (4) Yellow

23. When the circular scale of a screw gauge completes 2 rotations, it covers 1 mm over the pitch scale. The total number of the circular scale divisions is 50. The least count of the screw gauge in metre is:

(1) 10^{-5} (2) 10^{-2}
(3) 10^{-3} (4) 10^{-4}

24. A beam of light is incident vertically on a glass slab of thickness 1 cm, and refractive index 1.5. A fraction 'A' is reflected from the front surface while another fraction 'B' enters the slab and emerges after reflection from the back surface. Time delay between them is:

(1) 5×10^{-10} s (2) 10^{-11} s
(3) 5×10^{-11} s (4) 10^{-10} s

25. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A standing bus suddenly accelerates. If there were no friction between the feet of a passenger and the floor of the bus, the passenger would move back.

Reason (R): In the absence of friction, the floor of the bus would slip forward under the feet of the passenger.

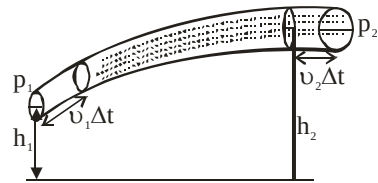
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) (A) is false but (R) is true
(2) Both (A) and (R) are true and (R) is the correct explanation of (A)
(3) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(4) (A) is true but (R) is false

26. A gas undergoes an isothermal process. The specific heat capacity of the gas in the process is:

(1) 0.5 (2) zero
(3) 1 (4) infinity

27. A fluid of density ρ is flowing in a pipe of varying cross-sectional area as shown in the figure. The Bernoulli's equation for the motion becomes:



(1) $p + \frac{1}{2}\rho v^2 = \text{constant}$

(2) $\frac{1}{2}\rho v^2 + \rho gh = \text{constant}$

(3) $p + \rho gh = \text{constant}$

(4) $p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$

28. At some instant, the number of radioactive atoms in a sample is N_0 and after time 't' the number decreases to N. It is found that the graphical representation 'ln N' versus 't' along the y and x axes respectively is a straight line. Then the slope of this line is:

(1) $-\lambda$ (2) λ^{-1}
(3) $-\lambda^{-1}$ (4) λ

29. A particle of mass 4 M kg at rest splits into two particles of mass M and 3 M. The ratio of the kinetic energies of mass M and 3M would be:

(1) 1 : 4 (2) 1 : 1
(3) 1 : 3 (4) 3 : 1

30. A hollow metal sphere of radius R is given '+Q' charge to its outer surface. The electric potential at a distance $\frac{R}{3}$ from the centre of the sphere will be:

(1) $\frac{3}{4\pi\epsilon_0} \frac{Q}{R}$

(2) $\frac{1}{4\pi\epsilon_0} \frac{Q}{3R}$

(3) $\frac{1}{4\pi\epsilon_0} \frac{Q}{R}$

(4) $\frac{1}{4\pi\epsilon_0} \frac{Q}{9R}$

31. Rain is falling vertically downward with a speed of 35 m/s. Wind starts blowing after some time with a speed of 12 m/s in East to West direction. The direction in which a boy standing at the place should hold his umbrella is:

- (1) $\tan^{-1}\left(\frac{12}{37}\right)$ w.r.t. wind
- (2) $\tan^{-1}\left(\frac{12}{35}\right)$ w.r.t. rain
- (3) $\tan^{-1}\left(\frac{12}{35}\right)$ w.r.t. wind
- (4) $\tan^{-1}\left(\frac{12}{37}\right)$ w.r.t. rain

32. The ratio of the moments of inertia of two spheres about their diameter and having same mass and their radii in the ratio 1 : 2 is:

- (1) 4 : 1
- (2) 1 : 2
- (3) 1 : 4
- (4) 2 : 1

33. A string is wrapped along the rim of a wheel of moment of inertia 0.10 kg-m^2 and radius 10 cm. If the string is now pulled by a force 10 N, then the wheel starts to rotate about its axis from rest. The angular velocity of the wheel after 2 seconds is:

- (1) 80 rad/s
- (2) 10 rad/s
- (3) 20 rad/s
- (4) 40 rad/s

34. Assuming the earth to be a sphere of uniform density, its acceleration due to gravity acting on a body:

- (1) increases with increasing depth
- (2) is independent of the mass of the earth
- (3) is independent of the mass of the body
- (4) increases with increasing altitude

35. If λ_x , λ_i , λ_M and λ_γ are the wavelengths of X-rays, infrared rays, microwaves and γ rays respectively, then:

- (1) $\lambda_M < \lambda_i < \lambda_x < \lambda_\gamma$
- (2) $\lambda_x < \lambda_\gamma < \lambda_M < \lambda_i$
- (3) $\lambda_x < \lambda_i < \lambda_\gamma < \lambda_M$
- (4) $\lambda_\gamma < \lambda_x < \lambda_i < \lambda_M$

Section-B (Physics)

36. The fraction of the original number of radioactive atoms that disintegrates (decays) during the average life time of a radioactive substance will be :

- (1) $\frac{1}{1+e}$
- (2) $\frac{e-1}{e+1}$
- (3) $\frac{e-1}{e}$
- (4) $\frac{1}{e}$

37. Three capacitors, each of capacitance $0.3 \mu\text{F}$ are connected in parallel. This combination is connected with another capacitor of capacitance $0.1 \mu\text{F}$ in series. Then the equivalent capacitance of the combination is:

- (1) $0.09 \mu\text{F}$
- (2) $0.1 \mu\text{F}$
- (3) $0.01 \mu\text{F}$
- (4) $0.9 \mu\text{F}$

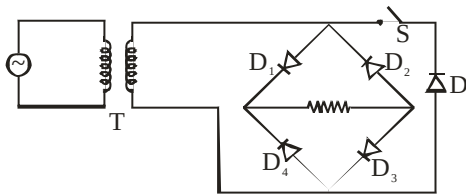
38. Given below are two statements:

Statement-I: The magnetic field of circular current loop at very far away point on the axial line varies with distance as like that of a magnetic dipole.

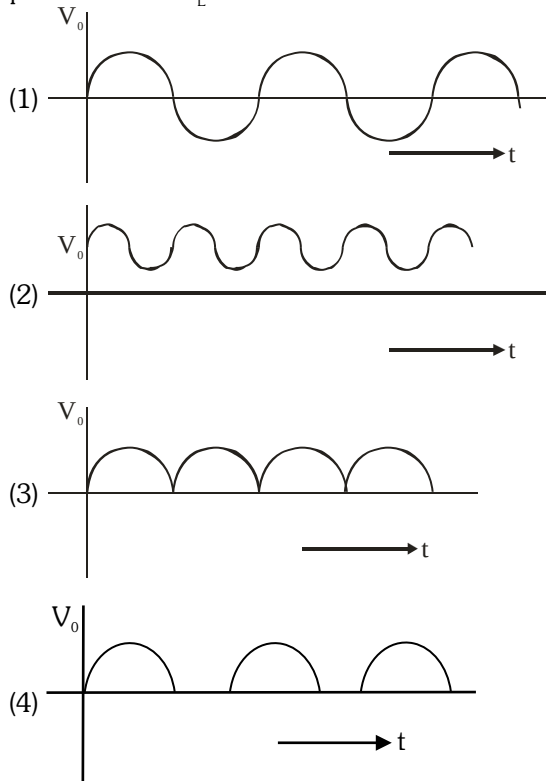
Statement-II: The magnetic field due to magnetic dipole varies inversely with the square of the distance from the centre on the axial line. In light of above statements, choose the most appropriate answer from the options given below:

- (1) **Statement-I** is incorrect and **Statement-II** is correct.
- (2) Both **Statement-I** and **Statement-II** are correct.
- (3) Both **Statement-I** and **Statement-II** are incorrect.
- (4) **Statement-I** is correct and **Statement-II** is incorrect.

39.



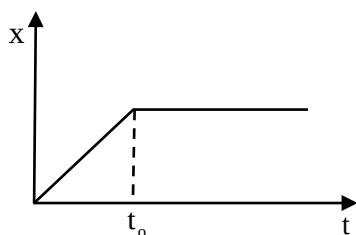
The circuit represents a full wave bridge rectifier when switch S is open. The output voltage (V_0) pattern across R_L when S is closed is:



40. When a particle with charge $+q$ is thrown with an initial velocity v towards another stationary charge $+Q$, it is repelled back after reaching the nearest distance r from $+Q$. The closest distance that it can reach if it is thrown with initial velocity $2v$, is:

- (1) $\frac{r}{2}$ (2) $\frac{r}{16}$ (3) $\frac{r}{8}$ (4) $\frac{r}{4}$

41. The figure given below shows the displacement and time, (x - t) graph of particle moving along a straight line:



The correct statement, about the motion of the particle is:

- (1) The particle is accelerated throughout its motion.
(2) The particle is accelerated continuously for time t_0 then moves with constant velocity.
(3) The particle is at rest.
(4) The particle moves at constant velocity up to a time t_0 and then stops.

42. The temperature at which the rms speed of atoms in neon gas is equal to the rms speed of hydrogen molecules at 15°C is:

(Atomic mass of neon = 20.2 u , molecular mass of $\text{H}_2 = 2 \text{ u}$)

- (1) 2.9 K (2) $0.15 \times 10^3 \text{ K}$
(3) $0.29 \times 10^3 \text{ K}$ (4) $2.9 \times 10^3 \text{ K}$

43. Air is pushed carefully into a soap bubble of radius r to double its radius. If the surface tension of the soap solution is T , then work done in the process is:

- (1) $24 \pi r^2 T$ (2) $4 \pi r^2 T$
(3) $8 \pi r^2 T$ (4) $12 \pi r^2 T$

44. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

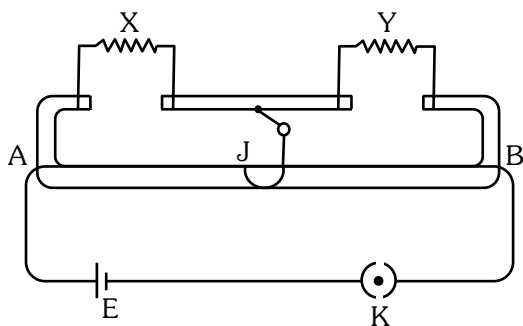
Assertion (A): Gauss's law for magnetism states that the net magnetic flux through any closed surface is zero.

Reason (R): The magnetic monopoles do not exist. North and South poles occur in pairs, allowing vanishing net magnetic flux through the surface.

In the light of the above statement, choose the most appropriate answer from the options given below:

- (1) **(A)** is false but **(R)** is true
(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
(3) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**
(4) **(A)** is true but **(R)** is false

45. A monochromatic light of frequency 500 THz is incident on the slits of a Young's double slit experiment. If the distance between the slits is 0.2 mm and the screen is placed at a distance 1 m from the slits, the width of 10 fringes will be: [THz = 10^{12} Hz]
- (1) 15 mm
 - (2) 30 mm
 - (3) 3 mm
 - (4) 1.5 mm
46. An a.c. source given by $V = V_m \sin \omega t$ is connected to a pure inductor L in a circuit and I_m is the peak value of the ac current. The instantaneous power supplied to the inductor is:
- (1) $-\frac{V_m I_m}{2} \sin(2\omega t)$
 - (2) $V_m I_m \sin^2(\omega t)$
 - (3) $-V_m I_m \sin^2(\omega t)$
 - (4) $\frac{V_m I_m}{2} \sin(2\omega t)$
47. In a metre bridge experiment, the null point is at a distance of 30 cm from A. If a resistance of 16Ω is connected in parallel with resistance Y , the null point occurs at 50 cm from A. The value of the resistance Y is:



- (1) $\frac{40}{3} \Omega$
- (2) $\frac{64}{3} \Omega$
- (3) $\frac{48}{3} \Omega$
- (4) $\frac{112}{3} \Omega$

48. Two planets are in a circular orbit of radius R and $4R$ about a star. At a specific time, the two planets and the star are in a straight line. If the period of the closest planet is T , then the star and planets will again be in a straight line after a minimum time:

- (1) $(4)^{\frac{1}{3}} T$
- (2) $2 T$
- (3) $8 T$
- (4) $(4)^2 T$

49. A string of length l is fixed at both ends and is vibrating in second harmonic. The amplitude at antinode is 2 mm. The amplitude of a particle at a distance $\frac{l}{8}$ from the fixed end is:

- (1) 4 mm
- (2) $\sqrt{2}$ mm
- (3) $2\sqrt{3}$ mm
- (4) $2\sqrt{2}$ mm

50. The determination of the value of acceleration due to gravity (g) by simple pendulum method employs the formula,

$$g = 4\pi^2 \frac{L}{T^2}$$

The expression for the relative error in the value of ' g ' is:

- (1) $\frac{\Delta g}{g} = 4\pi^2 \left[\frac{\Delta L}{L} - 2 \frac{\Delta T}{T} \right]$
- (2) $\frac{\Delta g}{g} = 4\pi^2 \left[\frac{\Delta L}{L} + 2 \frac{\Delta T}{T} \right]$
- (3) $\frac{\Delta g}{g} = \frac{\Delta L}{L} - 2 \frac{\Delta T}{T}$
- (4) $\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T}$

Section-A (Chemistry)

51. Match **List-I** with **List-II** :

List-I (Monomers)	List-II (Polymers)
(a) Caprolactam	(i) Bakelite
(b) Ethylene glycol and Benzene-1,4-dicarboxylic acid	(ii) Nylon 6,6
(c) Hexamethylenediamine and adipic acid	(iii) Nylon-6
(d) Phenol and Formaldehyde	(iv) Terylene

Choose the **correct answer** from the options given below :

- (1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
 (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

52. Which of the following is the correct statement ?

- (1) Al^+ is more stable than In^+
 (2) Ga^+ is more stable than Al^+
 (3) Al^+ is more stable than Ga^+
 (4) Ga^+ is more stable than In^+

53. Which one of the following statements is true about the structure of CO_3^{2-} ion ?

- (1) Out of the three C–O bonds, two are longer and one is shorter.
 (2) It has three sigma and three π -bonds.
 (3) All three C–O bonds are equal in length with a bond order in between 1 and 2.
 (4) It can be explained by considering sp^3 hybridization.

54. Which one of the following electrons in the ground state will have least amount of energy?

- (1) An electron in 2p orbital of carbon atom.
 (2) The electron of copper atom present in 4s orbital.
 (3) The outermost electron in sodium atom.
 (4) An electron in hydrogen atom.

55. Which of the following is not correct about postulates of kinetic molecular theory of gases?

- (1) Volume of the gas is due to the large number of molecules of the gas.
 (2) Average kinetic energy of molecules is directly proportional to the absolute temperature of the gas.
 (3) The molecules move randomly with different speeds in different directions.
 (4) Pressure of the gas is due to the collision of molecules against the walls of the container.

56. The work done when 1 mole of a gas expands reversibly and isothermally from pressure of 5 atm to 1 atm at 300 K is

(Given $\log 5 = 0.6989$ and $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)

- (1) 150 J
 (2) + 4014.6 J
 (3) –4014.6 J
 (4) zero J

57. LiF is sparingly soluble in water because it has

- (1) small electronegativity.
 (2) high lattice enthalpy.
 (3) low hydration enthalpy.
 (4) partial covalent character.

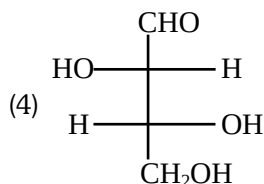
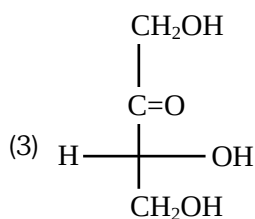
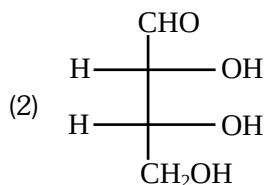
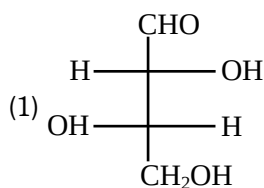
58. What is the **correct** order for boiling points of the following compounds ?

- (1) $\text{BiH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$
 (2) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
 (3) $\text{PH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
 (4) $\text{AsH}_3 > \text{PH}_3 > \text{NH}_3 > \text{SbH}_3 > \text{BiH}_3$

59. A ferromagnetic substance becomes a permanent magnet when it is placed in a magnetic field because

- (1) domains are not affected by magnetic field.
 (2) domains get randomly oriented.
 (3) all the domains get oriented in the direction of magnetic field.
 (4) all the domains get oriented in the direction opposite to the direction of magnetic field.

60. Which one is not a D-sugar ?



61. Match **List-I** with **List-II** :

List-I	List-II
(Amines)	(pK_b values)
(a) N-methylmethanamine	(i) 9.30
(b) Ammonia	(ii) 9.38
(c) N-methylaniline	(iii) 4.75
(d) Benzenamine	(iv) 3.27

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 (3) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
 (4) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

62. The number of bridging carbonyl groups in [Co₂(CO)₈] and [Mn₂(CO)₁₀], respectively are

- (1) 2 and 2
 (2) 2 and 4
 (3) 0 and 2
 (4) 2 and 0

63. Match **List-I** with **List-II** :

List-I	List-II
(a) Element which exhibits +3 oxidation state only	(i) Mn
(b) Element which exhibits more number of oxidation states	(ii) Zn
(c) Element which is a reducing agent in its +2 oxidation state	(iii) Sc
(d) Element which is not considered as a transition element	(iv) Cr

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
 (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

64. Given below are two statements :

Statement - I : The product of reaction of phenol with bromine depends on the nature of solvent.

Statement - II : Reaction of phenol with bromine in CHCl₃ gives monosubstituted bromo derivative whereas reaction of phenol with bromine water yields trisubstituted bromo derivative of phenol.

In light of the above statements, choose the most appropriate answer from the options given below:

- (1) **Statement-I** is incorrect and **Statement-II** is correct
 (2) Both **Statement-I** and **Statement-II** are correct
 (3) Both **Statement-I** and **Statement-II** are incorrect
 (4) **Statement-I** is correct and **Statement-II** is incorrect

65. Match List-I with List-II :

List-I (Example of Colloidal Systems)	List-II (Nature of dispersion medium and dispersed phase)
(a) Insecticide spray	(i) Dispersion medium-liquid Dispersed phase - solid
(b) Whipped Cream	(ii) Dispersion medium – gas Dispersed phase – liquid
(c) Paint	(iii) Dispersion medium– liquid Dispersed phase – liquid
(d) Hair Cream	(iv) Dispersion medium– liquid Dispersed phase - gas

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 (2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)
 (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

66. The solubility product of BaSO_4 in water is 1.5×10^{-9} . The molar solubility of BaSO_4 in 0.1 M solution of $\text{Ba}(\text{NO}_3)_2$ is

- (1) 0.5×10^{-8} M
 (2) 1.5×10^{-8} M
 (3) 1.0×10^{-8} M
 (4) 2.0×10^{-8} M

67. The correct order of acid strength of the following molecules is

- (1) $\text{C}_2\text{H}_5\text{OH} < \text{H}_2\text{O} < \text{CH}_3\text{COOH} < \text{C}_6\text{H}_5\text{OH}$
 (2) $\text{C}_2\text{H}_5\text{OH} < \text{H}_2\text{O} < \text{C}_6\text{H}_5\text{OH} < \text{CH}_3\text{COOH}$
 (3) $\text{H}_2\text{O} < \text{C}_2\text{H}_5\text{OH} < \text{C}_6\text{H}_5\text{OH} < \text{CH}_3\text{COOH}$
 (4) $\text{H}_2\text{O} < \text{C}_6\text{H}_5\text{OH} < \text{C}_2\text{H}_5\text{OH} < \text{CH}_3\text{COOH}$

68. Which one of the following is the **correct** order of decreasing bond enthalpies for the given species ?

- (1) $\text{N}_2 > \text{O}_2 > \text{O}_2^{2-} > \text{O}_2^-$
 (2) $\text{N}_2 > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (3) $\text{O}_2 > \text{N}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (4) $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{N}_2$

69. The correct reaction among the following is

- (1) $\text{C}_6\text{H}_5\text{Br} + \text{C}_6\text{H}_5\text{Br} \xrightarrow[\text{dry ether}]{\text{Na}} \text{C}_6\text{H}_5\text{C}_6\text{H}_5$
 (2) $2\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{dry ether}]{\text{Mg}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 (3) $2\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{Ethanol}]{\text{Na}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
 (4) $\text{CH}_3\text{CH}_2\text{Br} + 2\text{C}_6\text{H}_5\text{Br} \xrightarrow[\text{dry ether}]{\text{Na}} \text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$

70. Given below are two statements :

Statement-I : Aldehydes and ketones having at least one α -hydrogen undergo aldol condensation in the presence of dilute alkali as catalyst.

Statement-II : When aldol condensation is carried out between two different aldehydes, it is called cross aldol condensation. Ketones do not give this reaction.

In light of the above statements, choose the most appropriate answer from the options given below:

- (1) **Statement-I** is incorrect and **Statement-II** is correct.
 (2) Both **Statement-I** and **Statement-II** are correct.
 (3) Both **Statement-I** and **Statement-II** are incorrect.
 (4) **Statement-I** is correct and **Statement-II** is incorrect.

71. The **incorrect statement** among the following regarding food preservatives is

- (1) antioxidants used in wine are SO_2 and sulphites.
- (2) antioxidants help in preserving the food longer.
- (3) antioxidants react with oxygen with a slower rate than food.
- (4) sorbic acid and propanoic acid are good food preservatives.

72. One mole of sugar is dissolved in three moles of water at 298 K. The relative lowering of vapour pressure is

- (1) 0.20
- (2) 0.50
- (3) 0.33
- (4) 0.25

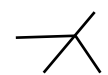
73. Match **List-I** with **List-II** :

List-I Elements	List-II Atomic radii (pm)
(a) O	(i) 88
(b) C	(ii) 74
(c) B	(iii) 66
(d) N	(iv) 77

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
- (3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

74. Match **List-I** with **List-II** :

List-I (Compound)	List-II (Boiling Point in K)
(a) 	(i) 300.9
(b) 	(ii) 282.5
(c) 	(iii) 309.1
(d) 	(iv) 341.9

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (4) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

75. Alkali metals, though white, impart a characteristic colour to the flame because of

- (1) oxidation of metal in the presence of flame.
- (2) excitation of valence electrons.
- (3) gain of electrons,
- (4) excitation and coming back of valence electrons to the original level.

76. Identify the set from the following sets in which all species can exhibit disproportionation reactions.

- (1) Cl_2 , ClO_2^- , ClO_3^- , S_8
- (2) ClO_4^- , ClO^- , ClO_2^- , F_2
- (3) ClO_3^- , ClO_4^- , H_2O_2 , ClO^-
- (4) ClO_2^- , ClO_3^- , ClO_4^- , Cl_2

77. For the reaction, $2\text{A} \rightarrow \text{B}$. rate = $k[\text{A}]^2$.

If concentration of reactant is doubled, then the

- (a) rate of reaction will be doubled.
- (b) rate constant will remain unchanged, however rate of reaction is directly proportional to the rate constant.
- (c) rate constant will change since rate of reaction and rate constant are directly proportional to each other.
- (d) rate of reaction will increase by four times.

Identify the set of correct statements :

Choose the **correct answer** from the options given below :

- (1) (a), (b) only
- (2) (b), (d) only
- (3) (c), (d) only
- (4) (a), (c) only

78. Match List-I with List-II :

List-I	List-II
(a) Separation of aniline-water mixture	(i) Fractional distillation
(b) Separation of aniline-chloroform mixture	(ii) Distillation under reduced pressure
(c) Separation of glycerol from spent-lye	(iii) Distillation
(d) Separation of different fractions of crude oil	(iv) Steam distillation

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
- (2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
- (3) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (4) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

79. Match List-I with List-II :

List-I	List-II
(a) 4.48 litres of O ₂ at STP	(i) 0.2 moles
(b) 12.022×10^{22} molecules of H ₂ O	(ii) 12.044×10^{23} molecules
(c) 96 g of O ₂	(iii) 6.4 g
(d) 88 g of CO ₂	(iv) 67.2 litres at STP

(Given – Molar volume of a gas at STP = 22.4 L)

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

80. Match List-I with List-II :

List-I (Commercial name)	List-II (Chemical name)
(a) Calgon	(i) Sodium aluminium silicate (hydrated)
(b) Permutit	(ii) Sodium carbonate
(c) Soap	(iii) Sodium hexameta-phosphate
(d) Washing soda	(iv) Sodium stearate

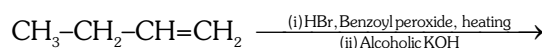
Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
- (4) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

81. Chlorine shows the bleaching action in the presence of moisture due to the formation of

- (1) HOClO
- (2) H₂O₂
- (3) O
- (4) HOCl

82. The product formed in the following reaction sequence is



- (1) CH₃–CH=CH–CH₃
- (2) CH₃–CH₂–CH=CH₂
- (3) CH₃–CH₂–CH₂–CH₂–OH
- (4) CH₃–CH₂–CH(OH)–CH₃

83. The three cells with their E°_(cell) values are given below :

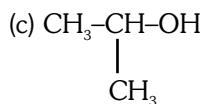
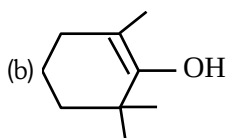
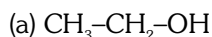
Cells	E° _(cell) / V
(a) Fe Fe ²⁺ Fe ³⁺ Fe	0.404
(b) Fe Fe ²⁺ Fe ³⁺ , Fe ²⁺ Pt	1.211
(c) Fe Fe ³⁺ Fe ³⁺ , Fe ²⁺ Pt	0.807

The standard Gibbs free energy change values for three cells are, respectively

(F represents charge on 1 mole of electrons)

- (1) +2.424 F, +2.422 F, +2.421 F
- (2) –0.808 F, –2.422 F, –2.421 F
- (3) –2.424 F, –2.422 F, –2.421 F
- (4) –1.212 F, –1.211 F, –0.807 F

84. The increasing order of reactivity of the following compounds towards acid catalysed dehydration is

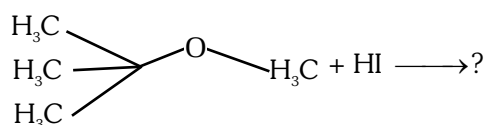


Choose the **correct answer** from the options given below :

- (1) (b) < (a) < (c) < (d)
 (2) (a) < (c) < (d) < (b)
 (3) (c) < (a) < (b) < (d)
 (4) (b) < (c) < (a) < (d)
85. The chain length of silicones can be controlled by adding
- (1) SiCl_4 (2) $(\text{CH}_3)_2\text{SiCl}_2$
 (3) $(\text{CH}_3)_3\text{SiCl}$ (4) CH_3SiCl_3

Section-B (Chemistry)

86. The major products formed in the following reaction are

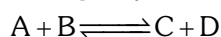


- (1)
 (2)
 (3)
 (4)

87. A student collected samples from two water bodies A and B in a metro city. The biochemical oxygen demand for 'A' is 3 ppm while for B is found to be 18 ppm. Which one of the following is **true**?

- (1) Both A and B are polluted.
 (2) A is clean but B is polluted.
 (3) A is polluted but B is clean.
 (4) Both A and B are clean.

88. Consider the following reaction taking place in 1 L capacity container at 300 K.



If one mole each of A and B are present initially and at equilibrium 0.7 mol of C is formed, then equilibrium constant (K_c) for the reaction is

- (1) 1.2 (2) 6.2
 (3) 5.4 (4) 9.7

89. Λ_m° for NaCl, HCl and CH_3COONa are 126.4, 425.9 and $91.05 \text{ S cm}^2 \text{ mol}^{-1}$ respectively. If conductivity of $0.001028 \text{ mol L}^{-1}$ acetic acid solution is $4.95 \times 10^{-5} \text{ S cm}^{-1}$, find the degree of dissociation of the acetic acid solution

- (1) 1.00 (2) 0.1233
 (3) 1.233 (4) 0.01233

90. The compound obtained by addition of water to an alkyne having more than two carbons, in presence of HgSO_4 and dilute H_2SO_4 at 333 K is

- (1) an aldehyde (2) an alcohol
 (3) a ketone (4) a vicinal diol

91. Given below are two statements :

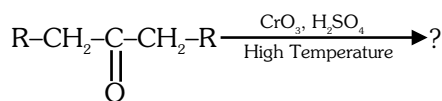
Statement-I : The Ellingham diagram provides an idea about the feasibility of a reaction.

Statement-II : The Ellingham explains the rate of the reduction reactions.

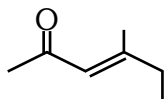
In light of above statements, choose the most **appropriate** answer from the options given below

- (1) **Statement-I** is incorrect and **Statement-II** is correct.
 (2) Both **Statement-I** and **Statement-II** are correct.
 (3) Both **Statement-I** and **Statement-II** are incorrect.
 (4) **Statement-I** is correct and **Statement-II** is incorrect.

92. The product(s) formed from the following reaction is/are

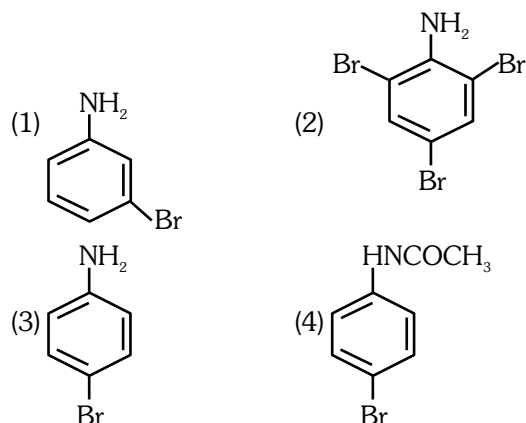
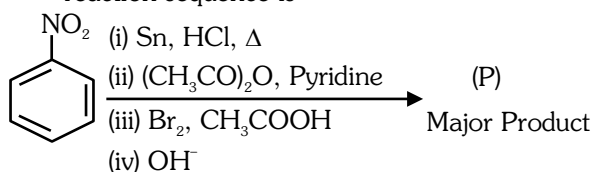


- (1) RCH_2COOH only
 (2) $\text{R}-\text{CH}_2-\underset{\text{COOH}}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{R}$ only
 (3) RCOOH and RCH_2COOH
 (4) RCOOH only
93. Identify the pair of Lanthanoides with one strong oxidant and one strong reductant.
 (1) Yb(II), Eu(II) (2) Eu(IV), Lu(III)
 (3) Ce(IV), Eu(II) (4) Ce(IV), Tb(IV)
94. A monochromatic infrared range finder of power 1 mW emits photons with wavelength 1000 nm in 0.1 second. The number of photons emitted in 0.1 second is
 (Given : $h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ ms⁻¹, Avogadro number = 6.022×10^{23})
 (1) 5×10^{14} (2) 30×10^{34}
 (3) 5×10^{11} (4) 30×10^{37}
95. The correct IUPAC name of the following compound is :-



- (1) 4-methylhex-3-en-2-one
 (2) 4-ethylpent-3-en-2-one
 (3) 3-methylhex-3-en-4-one
 (4) 2-ethylhex-3-en-4-one
96. At 300 K, 250 mL of gas A at 1 bar pressure is mixed with 500 mL of gas B at 2 bar pressure in a 1.0 L flask. Gas A does not react with gas B. The final pressure of the mixture is :
 (1) 2.15 bar (2) 2.50 bar
 (3) 1.25 bar (4) 1.00 bar

97. The major product (P) formed in the following reaction sequence is



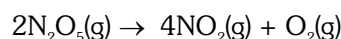
98. Which one of the following is the correct order of spin-only magnetic moment for the given complexes?

- (1) $[\text{Fe}(\text{CN})_6]^{3-} > [\text{Co}(\text{H}_2\text{O})_6]^{2+} > [\text{MnCl}_6]^{3-}$
 (2) $[\text{MnCl}_6]^{3-} > [\text{Fe}(\text{CN})_6]^{3-} > [\text{Co}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{MnCl}_6]^{3-} > [\text{Co}(\text{H}_2\text{O})_6]^{2+} > [\text{Fe}(\text{CN})_6]^{3-}$
 (4) $[\text{Co}(\text{H}_2\text{O})_6]^{2+} > [\text{MnCl}_6]^{3-} > [\text{Fe}(\text{CN})_6]^{3-}$

99. Which statement among the following is not correct?

- (1) Ferrimagnetism arises due to the alignment of magnetic moments of the domains in the substance in parallel and anti-parallel directions in unequal numbers.
 (2) Replacing some silicon atoms by boron atoms in crystal of silicon produces p-type semiconductor.
 (3) Replacing some germanium atoms by phosphorus atoms in crystal of germanium produces n-type semiconductor.
 (4) When conduction band and valence band overlap, a semiconductor is obtained.

100. The plot of $\ln k$ vs $\frac{1}{T}$ for the following reaction



gives a straight line with the slope of line equal to -1.0×10^4 K. Activation energy for the reaction in J mol⁻¹ is

(Given $R = 8.3$ J K⁻¹ mol⁻¹)

- (1) 4.0×10^{-2} (2) 8.3×10^{-4}
 (3) 8.3×10^4 (4) 4.0×10^2

Section-A (Biology : Botany)

- 101.** Which one of the following structures is haploid in its ploidy level?
- Microspore Mother Cell
 - Protonemal cell of a moss
 - Primary endosperm nucleus in dicot
 - Primary Endospore Nucleus
- 102.** Select the **correct** statement related to the activity of cork cambium.
- The cork differentiated from cork cambium, is impervious to water due to deposition of tannins and resins.
 - Cuts the cells only on the outer side.
 - Cuts the cells on inner as well as outer side.
 - The outer cells differentiate into phelloderm.
- 103.** Which one of the following process is responsible for the release of N_2 in the atmosphere?
- Ammonification
 - Denitrification
 - Biological nitrogen fixation
 - Industrial Nitrogen fixation
- 104.** DNA replication is semi-conservative in nature was experimentally proved in eukaryotes by :
- Macleod and McCarty
 - Meselson and Stahl
 - Talyor and his colleagues
 - Hershey and Chase
- 105.** High dose of UV-B causes inflammation of cornea and is called as :
- Colour-blindness
 - Evening-blindness
 - Snow-blindness
 - UV-blindness

106. Match List - I with List - II**List - I**

- ETS complex-I
- ETS complex-II
- ETS complex-III
- ETS complex-IV

List - II

- Cyt bc_1
- Cyt a, a_3 and 2 copper centres
- NADH dehydrogenase
- Ubiquinone and FADH dehydrogenase

Choose the **correct answer** from the options given below:

- (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
- (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
- (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

- 107.** The process of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration is referred as :

- Emigration
- Immigration
- Competition
- Association

- 108.** Which of the following come under the "Evil Quartet"?

- Habitat loss and fragmentation
- Over-exploitation
- Alien species invasion
- Mortality
- Competition

Choose the **correct** answer from the options given below :

- (b), (c) and (d)
- (a), (b) and (c)
- (a), (b) and (d)
- (a), (c) and (d)

- 109.** Given below are two statements :

Statement - I : Cellulose is a polymeric polysaccharide.

Statement - II : The building blocks of cellulose are glucose molecules.

In the light of the above statements, choose the **correct** answer from the options given below :

- Statement - I** is incorrect but **Statement - II** is correct
- Both **Statement - I** and **Statement - II** are correct
- Both **Statement - I** and **Statement - II** are incorrect
- Statement - I** is correct but **Statement - II** is incorrect

- 110.** The living differentiated cells, that lost the capacity to divide anymore, can regain the capacity of division under certain conditions. This phenomenon is termed as :

- (1) Maturation (2) Differentiation
(3) Dedifferentiation (4) Redifferentiation

111. Match List - I with List - II

List-I

(a) Haemophilia

(b) Down's Syndrome

(c) Phenylketonuria

(d) Klinefelter's Syndrome

List-II

(i) Inborn error of metabolism which lacks an enzyme that converts phenylalanine into tyrosine.

(ii) Sex-linked recessive disorder, defect in blood coagulation.

(iii) Presence of additional copy of X-chromosome (44+XXY)

(iv) Additional copy of chromosome number 21

Choose the **correct answer** from the options given below:

- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(2) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
(4) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

- 112.** Identify the cytochrome which acts as a mobile carrier for the transfer of electrons between complex III and IV?

- (1) Cytochrome a_3
(2) Cytochrome $b c_1$
(3) Cytochrome c
(4) Cytochrome a

- 113.** Which one of the following is not a criterion of genetic material?

- (1) Should be able to express itself in the form of Mendelian character.
(2) Should be able to generate its replica.
(3) Should be stable chemically and structurally.
(4) Should not provide the scope for changes for evolution.

- 114.** Identify the **correct** statements related to the androecium in the flower.

- (a) The sterile stamens are called staminodes.
(b) When stamens are attached to petals they are called epipetalous.
(c) Monadelphly is seen in China-rose.
(d) Polyadelphly is seen in Pea.
(e) Variation in the length of anther filaments is seen in Mustard.

Choose the **correct** answer from the options given below :

- (1) (a), (b), (c) and (e) only
(2) (a), (b) and (c) only
(3) (b), (c) and (d) only
(4) (a), (c), (d) and (e) only

- 115.** Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The growth of multicellular organism is due to mitosis.

Reason (R) : Mitosis is also called as equational division and it offers genetic stability.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
(3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(4) **(A)** is correct but **(R)** is not correct

- 116.** Identify the correct set of statements with regard to properties of humus.

- (a) Highly resistant to microbial action.
(b) Dark coloured amorphous substance.
(c) End product of detritus food chain.
(d) Reservoir of nutrients.
(e) Undergoes decomposition very fast.

Choose the correct answer from the options given below :

- (1) (a), (b) and (e) only
(2) (a) and (b) only
(3) (a), (b) and (c) only
(4) (a), (b) and (d) only

- 117.** The products of light reaction in photosynthesis are:
 (1) ATP, NADPH, O_2 and H_2O
 (2) ATP, NADPH and H_2O
 (3) ATP, NADPH and CO_2
 (4) ATP, NADPH and O_2
- 118.** Which one of the following experiments of Frederick Griffith resulted in the discovery of bacterial transformation?
 (1) S-strain (heat killed) + R-strain (live) → injected in to Mice → Mice died
 (2) S-strain → injected in to Mice → Mice died
 (3) R-strain → injected in to Mice → Mice lived
 (4) S-strain (heat killed) → injected in to Mice → Mice lived
- 119.** Which hormone is used to induce immediate stomatal closure in leaves?
 (1) Gibberellin (2) Abscissic Acid
 (3) Auxin (4) Cytokinin
- 120.** Select the **correct** statements with respect to pleiotropism.
 (a) A gene is said to be pleiotropic if it affects more than one trait.
 (b) Phenylketonuria is an example of pleiotropy.
 (c) A condition where one gene has several alleles is referred to as pleiotropism.
 (d) A trait is said to be pleiotropic if several genes control it.
 Choose the **correct** answer from the options given below :
 (1) (a) and (d) only (2) (a), (b) and (c) only
 (3) (b), (c) and (d) only (4) (a) and (b) only
- 121.** Which of the following is the **correct** equation of exponential growth?
 (1) $N_t = N_0 e^{rpt}$ (2) $N_t = N_0 e^{rst}$
 (3) $N_t = N_0 e^{rt}$ (4) $N_t = N_0 e^{mt}$
- 122.** Which of the following plants possesses the placentation of ovules borne on central axis with no septa?
 (1) Pea (2) China-rose
 (3) Primrose (4) Lemon
- 123.** What will be the ploidy of endosperm of a seed produced after crossing tetraploid female plant with tetraploid male plant?
 (1) Hexaploid (2) Diploid
 (3) Triploid (4) Pentaploid
- 124.** Removal of apical dominance by decapitation is utilised for :
 (1) Early senescence
 (2) Hedge making
 (3) Preparing weed-free lawns
 (4) Suppressing the activity of intercalary meristem
- 125.** Which of the following statements about facilitated diffusion is **incorrect**?
 (1) Porins are involved in this process.
 (2) Movement of molecule occurs against the concentration gradient.
 (3) ATP is not required for this process,
 (4) Special proteins of the membrane help in this process.
- 126.** Species Area relationship is described by the following equation.
 $\log S = \log C + Z \log A$
 where Z is :
 (1) Species richness
 (2) Slope of the line
 (3) Y-intercept
 (4) Area
- 127.** Genetically engineered insulin for human is produced from :
 (1) *Pseudomonas putida*
 (2) *Bacillus thuringiensis*
 (3) *Rhizobium meliloti*
 (4) *Escherichia coli*
- 128.** Which of the following physical properties of water is/are responsible for providing water, the high tensile strength and high capillarity during ascent of sap in the plants?
 (1) Cohesion, adhesion and surface tension
 (2) Cohesion only
 (3) Adhesion and cohesion
 (4) Surface tension and cohesion
- 129.** Choose the mismatched pair of leaf character with its example :
 (1) Alternate phyllotaxy - China-rose
 (2) Leaf tendril - Pea
 (3) Opposite phyllotaxy - *Calotropis*
 (4) Palmately compound - *Alstonia*

130. In prophase I of Meiosis, chromosomes start pairing together and synapsis takes place. This process occurs during which of the following stage?

- (1) Diplotene (2) Leptotene
(3) Zygotene (4) Pachytene

131. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Restriction enzyme is a type of endonuclease.

Reason (R) : Restriction enzyme cuts the two strands of DNA at specific positions within the DNA.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
(3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(4) **(A)** is correct but **(R)** is not correct

132. Axillary buds are derived from the activity of:

- (1) Secondary meristem
(2) Apical meristem
(3) Intercalary meristem
(4) Lateral meristem

133. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A father will never pass the gene for haemophilia to his sons.

Reason (R) : Haemophilia is sex-linked (X-linked) recessive trait.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
(3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(4) **(A)** is correct but **(R)** is not correct

134. Match **List - I** with **List - II**

List - I

List - II

- (a) *Cedrus* (i) Pteridophyte
(b) *Adiantum* (ii) Gymnosperm
(c) *Sphagnum* (iii) Liverwort
(d) *Marchantia* (iv) Moss

Choose the correct answer from the options given below :

- (1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(3) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(4) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

135. Phloem sap in the plants mainly consists of:

- (1) fructose and sucrose
(2) glucose and water
(3) sucrose and water
(4) fructose and water

Section-B (Biology : Botany)

136. Which of the following bond is formed as a result of reaction of carboxyl group of one amino acid with amino group of other amino acid with elimination of water?

- (1) Hydrogen Bond
(2) Glycosidic Bond
(3) Peptide Bond
(4) Phosphodiester Bond

137. Which of the following was proved by girdling experiment?

- (1) Symplastic movement of water occurs through interconnected protoplast.
(2) Xylem is responsible for uptake of water.
(3) Phloem is responsible for translocation of food.
(4) Apoplastic movement of water occurs through intercellular spaces.

138. Identify the fungi which do not belong to the group of other fungi among the following.

- (1) Puffballs
(2) Mushrooms
(3) Bracket Fungi
(4) Sac-fungi

139. The ratio of carbon dioxide fixation between C_4 plants and C_3 plants is :

- (1) 2 : 3 (2) 1 : 1
(3) 1 : 2 (4) 2 : 1

140. Following crops have been extensively cultivated in CO₂ rich atmosphere for higher yield :

- (1) Carrots and Tomatoes
- (2) Wheat and Sugar beet
- (3) Tomatoes and Bell pepper
- (4) Sugar beet and Cabbage

141. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In rDNA technology non recombinants transformed bacteria grow on the medium containing ampicillin as well as medium containing tetracycline.

Reason (R) : Recombinant plasmids contain the foreign DNA or gene of interest.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
- (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (3) Both **(A)** and **(R)** are correct, but **(R)** is **not** the correct explanation of **(A)**
- (4) **(A)** is correct but **(R)** is not correct

142. The thickness of the ozone in a column of air from the ground to the top of the atmosphere is measured in terms of :

- (1) Swedberg units (2) Monomeric units
- (3) Dobson units (4) Balton units

143. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Semiconservative replication was experimentally proved by Mathew Meselson and Franklin Stahl (1958).

Reason (R) : Meselson and Stahl used radioactive isotope ¹⁵N and equilibrium density gradient centrifugation technique.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
- (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(4) **(A)** is correct but **(R)** is not correct

144. Which of the following are **not** correct regarding decomposition of wastes?

- (a) Low temperature inhibits decomposition.
- (b) Warm and moist environment favours the process.
- (c) The process is anaerobic.
- (d) It is slower if detritus is rich in proteins and carbohydrates.
- (e) Detritus is degraded into simpler inorganic substances by fungal and bacterial enzymes.

Choose the **correct** answer from the options given below :

- (1) (c), (a) and (d) only (2) (c) and (d) only
- (3) (c), (d) and (e) only (4) (b) and (c) only

145. Which of the following is not a character of collenchyma tissue?

- (1) They occur in layers below epidermis in dicotyledonous plants.
- (2) They consist of cells with thick corners due to cellulose deposition.
- (3) They are usually dead and without protoplasts.
- (4) They provide mechanical support to the growing part of the plant.

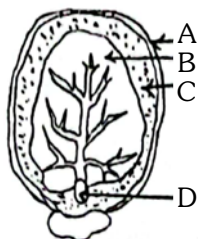
146. Assuming that fur colour of an animal is dark, range of colour shade and white. A cross is made between a male (AABBCC) with dark fur colour and a female (aabbcc) with white fur colour. What would be the fur colour of F₁ generation?

- (1) Range of colour shade
- (2) All dark colour
- (3) All white colour
- (4.) All intermediate colour

147. Which of the following pair of micronutrients would help in the light phase of photosynthesis to help in the reaction leading to oxygen evolution?

- (1) Manganese and Molybdenum
- (2) Molybdenum and Iron
- (3) Manganese and Chlorine
- (4) Zinc and Chlorine

148. Which of the following set represents the correct labelling of A, B, C and D with respect to the given diagram?



- | | |
|------------------|--------------|
| (1) A-Seed Coat, | B-Scutellum, |
| C-Endocarp, | D-Mesocarp |
| (2) A-Seed Coat, | B-Scutellum |
| C-Micropyle, | D-Endocarp |
| (3) A-Pericarp, | B-Coleoptile |
| C-Endosperm, | D-Scutellum |
| (4) A-Seed Coat, | B-Cotyledon, |
| C-Endosperm, | D-Hypocotyle |

149. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and formation of X shaped structures called chiasmata.

Reason (R) : In oocytes of some vertebrates, diplotene can last for months or years.

In the light of the above statements, choose the correct answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
- (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (3) Both **(A)** and **(R)** are correct but **(R)** not the correct explanation of **(A)**
- (4) **(A)** is correct but **(R)** is not correct

150. The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid of which of the following organism?

- (1) *Bacillus thuringiensis*
- (2) *Salmonella typhimurium*
- (3) *Agrobacterium tumefaciens*
- (4) *Escherichia coli*

Section - A (Biology : Zoology)

151. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A) : With the help of several ommatidia, a cockroach can perceive several images of an object, ie, mosaic vision.

Reason (R) : Mosaic vision gives more sensitivity but less resolution.

In the light of the above statements, choose the **most appropriate** answer from the option given below :

- (1) **(A)** is not correct but **(R)** is correct
- (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
- (4) **(A)** is correct but **(R)** is not correct

152. Inadequate supply of oxygen to heart muscles leads to a symptom of acute chest pain. This disorder of the circulatory system is identified as :

- (1) Cardiac arrest
- (2) Heart failure
- (3) Coronary Heart Disease
- (4) Angina pectoris

153. In *Drosophila*, the genes for color of body and color of eyes are situated on _____.

- (1) autosomes
- (2) Y-chromosome
- (3) X-chromosome
- (4) both the sex chromosomes

154. An intestinal hormone that stimulates the pancreas to release watery secretion that is rich in bicarbonate ions :

- (1) Gastric Inhibitory Peptide
- (2) Enterokinase
- (3) Secretin
- (4) Cholecystokinin

- 155.** The hormone releasing IUDs among the following are :
 (a) Multiload 375
 (b) LNG - 20
 (c) Progestasert
 (d) Lippe's loop
 (e) Vaults
 Choose the **most appropriate** answer from the options given below :
 (1) (c) and (e) only
 (2) (a) and (b) only
 (3) (b) and (c) only
 (4) (a) and (d) only
- 156.** Which of the following disorders represents decrease in respiratory surface due to damaged alveolar walls?
 (1) Bronchitis (2) Asthma
 (3) Emphysema (4) Hypocapnia
- 157.** Which of the following is/are vegetative propagule(s)?
 (a) Eyes of Potato
 (b) Zoospore of Chlamydomonas
 (c) Rhizome of Ginger (d) buds in Hydra
 (e) Bulbil of *Agave*
 Choose the **most appropriate** answer from the options given below :
 (1) (a), (c), (e) only (2) (a), (b), (c), (d) only
 (3) (b), (d) only (4) (e) only
- 158.** In a cell, the separation of DNA strands is brought about by the enzyme DNA helicase, whereas in PCR, the separation of DNA strands is due to :
 (1) Two sets of Primers
 (2) Taq DNA polymerase
 (3) Deoxynucleotides
 (4) High temperature
- 159.** Which of the following types of epithelium lines the walls of blood vessels?
 (1) Squamous epithelium (2) Cuboidal epithelium
 (3) Columnar epithelium (4) Ciliated epithelium
- 160.** Endemism refers to :
 (1) Species evenness
 (2) Species confined to that region
 (3) Species diversity
 (4) Species richness
- 161.** The term 'blue Revolution' is related with :
 (1) Development of water reservoirs
 (2) Honey and its by products
 (3) Fishery industry
 (4) Various crop plants and their by products
- 162.** The oocytes of some vertebrates get arrested for years or months in:
 (1) diplotene (2) Diakinesis
 (3) Metaphase - I (4) Telophase - I
- 163.** Which one of the following is **not** an Assisted Reproductive Technology (ART) used by childless couples to have children?
 (1) IUD (2) IVF
 (3) IUT (4) ZIFT
- 164.** A population with finite resources shows a logistic growth curve where the correct sequence of events will be :
 (1) Acceleration phase → Deceleration → Asymptote
 (2) Acceleration phase → Lag phase → Stationary phase
 (3) Lag phase → Acceleration phase → Deceleration → Asymptote
 (4) Stationary phase → Acceleration phase → Lag phase → Asymptote
- 165.** A low frequency recombination indicates that the genes are :
 (1) Located close to each other
 (2) Not linked
 (3) Present on different chromosomes
 (4) Located far apart from each other
- 166.** Arrange the following male sex accessory ducts in the correct sequence for the transport of sperms from the testes.
 (a) Epididymis (b) Ejaculatory duct
 (c) Vasa efferentia (d) Rete testis
 (e) Vas deferens
 Choose the **most appropriate** answer from the options given below :
 (1) (d), (c), (e), (a), (b)
 (2) (d), (c), (a), (e), (b)
 (3) (d), (e), (a), (c), (b)
 (4) (d), (a), (c), (e), (b)

167. In the regulation of respiration, a chemo-sensitive area adjacent to the rhythm centre in the medulla region of the brain, is highly sensitive to :

- (1) CO_2 (2) O_2
(3) N_2 (4) HCO_3^-

168. Chylomicrons are :

- (1) micro-sized lipid molecules
(2) protein coated fat globules
(3) spherical aggregates of fatty acids
(4) fat coated protein globules

169. Genetic Drift occurs due to :

- (1) Sudden population migration
(2) Continuous gene migration
(3) Mutation
(4) Natural selection

170. Immuno-suppressants are administered to burn-patients or during organ transplantation to suppress :

- (1) Cytokine storm
(2) Humoral immunity
(3) Cell-mediated immunity
(4) Innate immunity

171. Normal sleep-wake cycle in a human body is maintained by the secretion of :

- (1) Thymus gland
(2) Pineal gland
(3) Pituitary gland
(4) Thyroid gland

172. Air bladder is found in :

- (1) Aves
(2) Cyclostomata
(3) Chondrichthyes
(4) Osteichthyes

173. With respect to nucleosome, which of the following statements is **incorrect**?

- (1) Nucleosomes are seen as 'beads on string' under Electron Microscope
(2) DNA is wrapped around positively charged histone octamer to form nucleosome.
(3) Nucleosome is the repeating unit of chromatin
(4) Nucleosome contains 120 bp of DNA helix

174. In a reaction catalyzed by an enzyme, which of the following statements is **correct**?

- (1) Enzymes make transition from substrate to product more difficult.
(2) Enzymes increase the activation energy for formation of transition state.
(3) Enzyme-substrate complex formed during a reaction lasts for a very long time.
(4) Enzyme decrease the activation energy for formation of transition state.

175. Given below are two statements :

Statements-I : Membrane-bound organelles of the endomembrane system coordinate cellular functions.

Statement-II : Mitochondria and chloroplasts are not considered a part of the endomembrane system.

In the light of the above statements choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
(2) Both **Statement-I** and **Statement-II** are correct
(3) Both **Statement-I** and **Statement-II** are incorrect
(4) **Statement-I** is correct but **Statement-II** is incorrect

176. One of the strands of double stranded DNA has base composition as follows : 15% A, 15% T, 40% G and 30% C. What will be the percentage of these bases in the complementary strand?

- (1) 15% A, 10% T, 40% G, 15% C
(2) 15% A, 15% T, 40% G, 30% C
(3) 15% A, 40% T, 15% G, 30% C
(4) 15% A, 15% T, 30% G, 40% C

177. Given below are two statements:

Statement-I : Amphibians and reptiles have a 3-chambered heart with two atria and a single ventricle; and oviparous in nature.

Statement-II : Crocodiles possess a 4 chambered heart with two ventricles and two atria; and viviparous in nature.

In the light of the-above statements, choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
- (2) Both **Statement-I** and **Statement-II** are correct
- (3) Both **Statement-I** and **Statement-II** are incorrect
- (4) **Statement-I** is correct but **Statement-II** is incorrect

178. Herbarium, Botanical gardens, Museum, Zoological parks and Key are considered as-

- (1) Environmental aids
- (2) Pollution aids
- (3) Taxonomical aids
- (4) Trophic aids

179. Ear wax secreting cells have which type of epithelium?

- (1) Exocrine glandular epithelium
- (2) Compound epithelium
- (3) Endocrine glandular epithelium
- (4) Columnar epithelium

180. Select the **correct** sequence of events occurring during Prophase-I of Meiosis-I.

- (a) Nuclear envelope breakdown
- (b) Synaptonemal complex formation
- (c) Compaction of chromosomes
- (d) Terminalisation of chiasmata
- (e) Crossing over

Choose the **most appropriate** answer from the options given below :

- (1) (c) → (a) → (b) → (d) → (e)
- (2) (b) → (c) → (a) → (d) → (e)
- (3) (c) → (a) → (b) → (e) → (d)
- (4) (c) → (b) → (e) → (d) → (a)

181. Which of the following is a correct statement?

- (1) Z-lines anchor myosin (thick) filament to the ends of the sarcomere.
- (2) Sarcoplasmic reticulum stores acetylcholine.
- (3) Myosin cross bridges contain calcium binding sites.
- (4) Actin and regulatory proteins are located in thin filament.

182. How many Y-chromosomes are present in 2nd polar body in human beings?

- (1) 00
- (2) 23
- (3) 02
- (4) 01

183. Which one of the following hormones reduce the blood pressure?

- (1) Atrial Natriuretic factor
- (2) Aldosterone
- (3) Angiotensin-II
- (4) Antidiuretic hormone

184. Which of the following hormones are secreted in women only during pregnancy?

- (a) Relaxin
- (b) Oxytocin
- (c) hCG
- (d) hPL
- (e) Progesterone

Choose the **most appropriate** answer from the options given below :

- (1) (b) and (e) only
- (2) (b), (c) and (d) only
- (3) (a), (c) and (d) only
- (4) (c), (d) and (e) only

185. Bee-keeping helps to improve the yield of following crops EXCEPT _____.

- (1) Sunflower
- (2) Apple
- (3) Mustard
- (4) Jowar

Section - B (Biology : Zoology)

186. Match **List-I** with **List-II** regarding sensory organs in human.

- | | |
|--------------------|--------------------------|
| (a) Organ of corti | (i) Photo receptors |
| (b) Nasal mucosa | (ii) Gustatory receptors |
| (c) Taste buds | (iii) Auditory receptors |
| (d) Retina | (iv) Olfactory receptors |

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (2) (a)-(iv), (b)-(iii), (c)-(i) (d)-(ii)
- (3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (4) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

187. Given below are two statements :

Statement-I : Pyramid of energy is always upright and is the most efficient.

Statement-II : Pyramid of biomass in sea is generally inverted.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
- (2) Both **Statement-I** and **Statement-II** are correct
- (3) Both **Statement-I** and **Statement-II** are incorrect
- (4) **Statement-I** is correct but **Statement-II** is incorrect

188. Terrestrial adaptations necessitated the production of :

- (1) Lesser toxic nitrogenous wastes like urea and uric acid
- (2) Lesser toxic nitrogenous wastes like ammonia and urea
- (3) Highly toxic nitrogenous wastes like ammonia and urea
- (4) Highly toxic nitrogenous wastes like urea and uric acid

189. Which biological process leads to decrease in fish-eating bird population near a water body containing toxicants from industrial drainage?

- (1) Biochemical oxygen demand
- (2) Accelerated Eutrophication
- (3) Biomagnification
- (4) Algal bloom

190. What would be the proportions of light and hybrid density DNA molecule, respectively if Meselson and Stahl's experiment was continued for 60 minutes ?

- (1) 25%, 75%
- (2) 75%, 25%
- (3) 100%, 0%
- (4) 50%, 50%

191. Identify the properties of a good vector used in rDNA technology.

- (a) It should have origin of replication supporting high copy number.
- (b) It should have preferably more than '2' recognition sites.
- (c) the restriction sites in vector should be in the antibiotic resistant genes.
- (d) It should have suitable marker genes.
- (e) It should be easy to isolate and purify.

Choose the most appropriate answer from the options given below :

- (1) (c), (d) and (e) only
- (2) (a), (b) and (c) only
- (3) (a), (c), (d) and (e) only
- (4) (a), (c) and (e) only

192. Given below two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The nematode can not survive in a transgenic host which expresses specific interfering RNA.

Reason (R) : Nematode specific gene introduced in the host produces both sense and antisense complementary RNA which initiate RNA interference in the host cell.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
- (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
- (4) **(A)** is correct but **(R)** is not correct

193. Match **List - I** with **List - II**

List - I	List - II
(a) Adhering junctions	(i) Establish a barrier that prevents leakage of extracellular fluid across a layer of cells
(b) Tight junctions	(ii) Functions like rivets and fasten cells together into strong sheets
(c) Gap junctions	(iii) Pass information through neurotransmitters from one cell to another
(d) Synaptic junctions	(iv) Provide cytoplasmic channels from one cell to adjacent cell for communication

Choose the **correct answer** from the options given below:

- (1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
 (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
 (3) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

194. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In human beings. Insulin is synthesized as a pro-hormone which needs to be processed before it becomes fully mature and functional.

Reason (R) : The extra stretch of C-peptide is to be removed from A-peptide and B-peptide chain of insulin.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) **(A)** is not correct but **(R)** is correct
 (2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
 (3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
 (4) **(A)** is correct but **(R)** is not correct

195. Which one of the following features are **not** true for chordates?

- (a) Heart is dorsal.
 (b) Pharynx is perforated by gill slits.
 (c) Central nervous system is ventral, solid and single.
 (d) Post-anal tail is present.
 (e) Notochord is present.

Choose the **most appropriate** answer from the options given below :

- (1) (a), (d) and (c) only
 (2) (b) and (c) only
 (3) (a) and (c) only
 (4) (e) only

196. Match **List - I** with **List - II**

List - I	List - II
(a) Chromoplasts	(i) Proteins
(b) Amyloplasts	(ii) Oil and fats
(c) Elaioplasts	(iii) Starch
(d) Aleuroplasts	(iv) Carotene

Choose the **correct answer** from the options given below :

- (1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (2) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
 (3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
 (4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

197. Match **List - I** with **List - II**

List - I	List - II
(a) <i>Puccinia</i>	(i) Parasitic fungus on mustard
(b) <i>Neurospora</i>	(ii) Dead substrates
(c) Saprophytes	(iii) Wheat rust
(d) <i>Albugo</i>	(iv) Biochemical and Genetic work

Choose the **correct answer** from the options given below :

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
 (2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
 (3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
 (4) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

198. Select the **correct** match regarding adaptive radiation of Australian marsupials corresponding to placental mammals.

- (1) Tasmanian Wolf - Bobcat
- (2) Marsupial mouse - Mole
- (3) Spotted Cuscus - Lemur
- (4) Numbat - Flying Squirrel

199. Given below are two statements :

Statement-I : When an infected female *Anopheles* mosquito bites, it releases gametocytes of *Plasmodium* into the healthy person.

Statement-II : The female *Anopheles* mosquito takes up sporozoites of *Plasmodium* with blood meal from an infected person, suffering from malaria.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) **Statement-I** is incorrect but **Statement-II** is correct
- (2) Both **Statement-I** and **Statement-II** are correct
- (3) Both **Statement-I** and **Statement-II** are incorrect
- (4) **Statement-I** is correct but **Statement-II** is incorrect

200. If a DNA molecule is shortened by 25 base pairs, how many helical turns will be reduced from its structure?

- (1) 3
- (2) 2.5
- (3) 2
- (4) 1

ANSWER KEY

NEET(UG)-2022(Overseas)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	4	3	4	1	4	3	2	4	3	3	3	3	3	4
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	4	4	3	3	3	1	2	1	4	1	4	4	1	4	3
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	2	3	3	3	4	3	1	4	4	4	4	4	1	2	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	1	2	3	2	4	2	2	3	4	1	3	2	1	3	1
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	1	4	3	2	4	2	2	2	1	3	3	4	3	2	4
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	1	2	3	1	1	3	2	3	1	3	2	2	3	2	3
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	4	3	3	1	1	3	3	3	4	3	2	3	2	3	3
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	3	3	2	2	3	4	3	4	1	2	4	4	1	2	4
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	3	3	1	2	2	2	4	1	4	3	2	2	2	2	3
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	3	3	4	4	3	3	3	4	2	3	4	3	4	3	2
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	2	4	3	3	3	3	1	4	1	2	3	1	1	3	1
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	2	1	2	1	3	2	4	4	4	2	4	4	3	1	4
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	4	1	1	3	4	1	2	1	3	2	3	2	2	2	3
Que.	196	197	198	199	200										
Ans.	1	1	3	3	2										

HINT - SHEET

1. At $T_1 = 27^\circ\text{C} = 300\text{ K}$

$$\lambda_1 = \lambda$$

$$\text{and } T_2 = 927^\circ\text{C} = 1200\text{ K}$$

$$\lambda_2 = ?$$

$$\therefore \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \text{ and } E = \frac{3}{2} K_B T$$

$$\therefore \lambda \propto \frac{1}{\sqrt{T}}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\sqrt{T_2}}{\sqrt{T_1}} = \sqrt{\frac{1200}{300}} = 2$$

$$\lambda_2 = \frac{\lambda_1}{2} = \frac{\lambda}{2}$$

2. Since, $M = IA$

$$\therefore M_1 = M_2 \Rightarrow I_1 \pi r_1^2 = I_2 \pi r_2^2$$

$$\Rightarrow \frac{I_1}{I_2} = \left(\frac{r_2}{r_1} \right)^2 = \frac{4}{1}$$

3. $L = 4H$

$$\therefore \varepsilon = -L \frac{di}{dt} = -4 \frac{(2-4)}{1} = -4(-2) = 8\text{ V}$$

4. Wave is propagating along $-z$ direction

$$\vec{E} = 3\hat{j}v/m$$

$$\therefore B_0 = \frac{E_0}{C} = \frac{3}{3 \times 10^8}$$

$$= 10 \times 10^{-9} \hat{i} T = 10 \hat{i} nT$$

5. $\therefore A_V = A_{V_1} A_{V_2}$

$= 20 \times 20 = 400$

$\frac{v_{out}}{v_{in}} = A_V \Rightarrow v_{out} = A_V(v_{in})$

$= 400(0.01) = 4 \text{ V}$

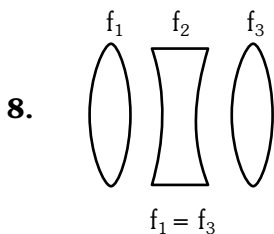
6. $\therefore I = I_0 \cos^2 \theta$

$2.5 = 10 \cos^2 \theta \Rightarrow \frac{25}{100} = \cos^2 \theta$

$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$

7. $v^2 - u^2 = 2gh$

$\Rightarrow h = \frac{v^2 - u^2}{2g} = \frac{3600 - 1600}{2 \times 10} = 100 \text{ m}$



$f_2 = -25 \text{ cm}$

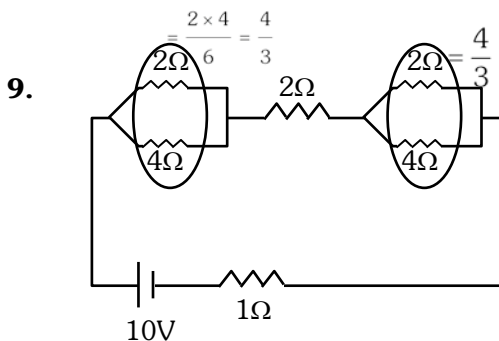
$f_1 = f_3 = 40 \text{ cm}$

$\therefore \frac{1}{f_{net}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$

and $P = \frac{1}{f_{net}(m)}$

$\therefore \frac{1}{f_{net}} = \frac{1}{40} - \frac{1}{25} + \frac{1}{40}$
 $= \frac{2}{40} - \frac{1}{25} = \frac{1}{100 \text{ cm}}$

$\therefore P = \frac{100}{100} = 1D$



$R_{eq} = \frac{4}{3} + 2 + \frac{4}{3} + 1 = \frac{8}{3} + 3 = \frac{17}{3} \Omega$

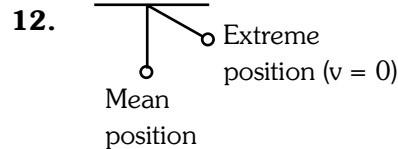
10. Viscosity of liquid decreases with increase of temp.

11. 1st and 2nd excited state of hydrogen corresponds to 2nd and 3rd orbital.

$n_1 = 2$ and $n_2 = 3$

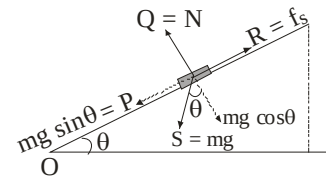
$\therefore L = \frac{nh}{2\pi}$

$\therefore \frac{L_1}{L_2} = \frac{n_1}{n_2} = \frac{2}{3}$



$\therefore$ At extreme position K.E. = 0

So, Total energy = Purely Potential Energy



13.

Q = Normal reaction

P = $mg \sin \theta$

R = Friction force

S = Weight (mg)

(a)-(ii), (b)-(iii), (c)-(iv) and (d)-(i)

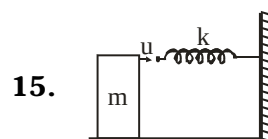
14. $\vec{E}$ at centre = 0

Because of symmetry when one charge is removed

$\vec{E} = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

($\therefore \vec{E}$ due to 10 charges will cancel and out each other

$\therefore E_{net}$ will be only because of 11th charge)



$\frac{1}{2} mu^2 = \frac{1}{2} kx^2$

$x = u \sqrt{\frac{m}{k}}$

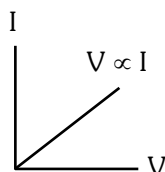
16. $y = \overline{(A+B)} + \overline{(A+B)}$

$= \overline{(A+B)} \cdot \overline{(A+B)}$ (By De'Morgans theorem)

$= (A+B) \cdot (A+B)$

$= A+B \Rightarrow \text{OR Gate}$

17. By Ohm's law : $V = IR \Rightarrow V \propto I$



B	B	R	O	Y	G	B	V	G	W
0	1	2	3	4	5	6	7	8	9
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Multiplier 1	10	10 ²	10 ³	10 ⁴	10 ⁵	10 ⁶	10 ⁷	10 ⁸	10 ⁹
Tolerance (%) : Gold, Silver, None									
	5	10	20						
(360 ± 18) kΩ									

19. $E = \frac{1}{2} LI^2$

$= \frac{1}{2} \times 10 \times 1^2$

$= \frac{10}{2} = 5 \text{ J}$

20. Dimensional formula of mutual inductance
 $= [ML^2T^{-2}A^{-2}]$

21. $\therefore F = q(\vec{v} \times \vec{B})$

Since $\vec{v}$ and $\vec{B}$ are parallel

$\therefore \theta = 0^\circ$

$F_m = Bqv \sin 0^\circ = 0$

$\therefore$ Original path

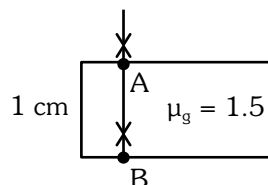
22. $\therefore$ Wavelength of blue, indigo and violet light is less than green light
 $\therefore$ Energy of blue, indigo and violet light is more than green light
Hence, violet light is capable of ejecting of photoelectron.

23. Pitch = $\frac{1\text{mm}}{2} = 0.5 \text{ mm}$

L.C. = $\frac{\text{Pitch}}{\text{No. of division of circular scale}}$

$= \frac{0.5 \times 10^{-3}}{50} = 10^{-5}$

24.



$v = \frac{s}{t}$ $\mu_{ga} = \frac{\mu_g}{\mu_a} = \frac{c}{v_g}$

$s = 2$ (Thickness of slab)

$t = \frac{s}{v_g} = \frac{(2 \times 10^{-2}) \times 3}{3 \times 10^8 \times 2}$

$\Rightarrow v_g = \frac{c\mu_a}{\mu_g} = \frac{2c}{3} = 10^{-10} \text{ s}$

25. $\therefore$ There is no friction between feet of passenger and floor of bus so passenger will not move back when seen from ground
So Assertion is false but reason is true.

26. $\therefore Q = ms \Delta T$

as $\Delta T = 0$

$s = Q/m\Delta T = \infty$

27. By using Bernoulli's equation

$p + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$

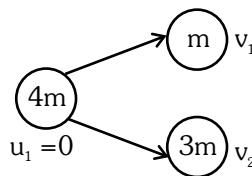
28. $N = N_0 e^{-\lambda t}$

$\ln N = \ln N_0 + \ln_e(e^{-\lambda t})$

$\ln N = -\lambda t + \ln N_0$

$\therefore$ Slope = $-\lambda$

29.



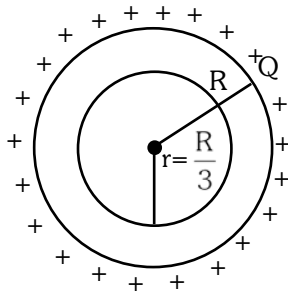
$\therefore F_{\text{ext}} = 0 \therefore \Delta P = 0$

$\Rightarrow P_1 + P_2 = 0 \Rightarrow P_1 = -P_2$

and $E = \frac{P^2}{2m}$

$\therefore E \propto \frac{1}{m} \Rightarrow \frac{E_1}{E_2} = \frac{3m}{m} = \frac{3}{1}$

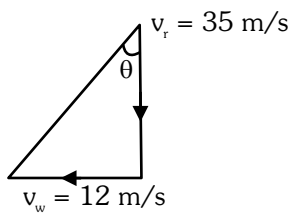
30.



Here potential inside sphere will be equal to potential on surface of sphere.

$$V = \frac{kQ}{R} = \frac{1}{4\pi\epsilon_0} \frac{Q}{R}$$

31.



$$\tan\theta = \frac{12}{35} \Rightarrow \theta = \tan^{-1}\left(\frac{12}{35}\right) \text{ w.r.t rain}$$

32. $\therefore I \propto mR^2$

$$\frac{m_1}{m_2} = \frac{1}{1}$$

$$\frac{R_1}{R_2} = \frac{1}{2}$$

$$\therefore \frac{I_1}{I_2} = \frac{m_1}{m_2} \times \left(\frac{R_1}{R_2}\right)^2$$

$$= 1 \times \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

33. $I = 0.1 \text{ Kg m}^2$

$$r = 10 \times 10^{-2} \text{ m}$$

$$F = 10 \text{ N}$$

$$t = 2 \text{ sec}$$

$$\therefore \tau = I\alpha = rF$$

$$\therefore \alpha = \frac{rF}{I} = \frac{10^{-1} \times 10}{10^{-1}} = 10 \text{ rads}^{-2}$$

$$\therefore \omega = \omega_0 + \alpha t$$

$$\therefore \omega = 0 + \alpha t$$

$$\Rightarrow \omega = 10 \times 2 = 20 \text{ rads}^{-1}$$

34. Acceleration due to gravity acting on a body is independent of the mass of the body.

$$\left(\because g = \frac{GM}{R^2}\right)$$

35.

R → Radio

m → Micro

I → Infrared

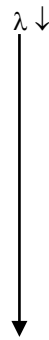
V → Visible

U → UV ray

X → X-ray

G → γ-ray

$$\lambda_m > \lambda_i > \lambda_x > \lambda_\gamma$$



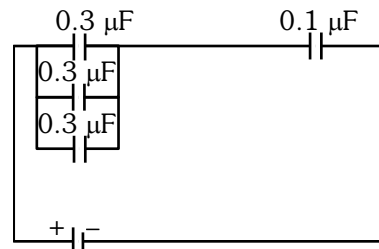
$$36. \frac{N_0 - N}{N_0} = \frac{N_0 - N_0 e^{-\lambda t}}{N_0} = 1 - e^{-\lambda t}$$

$$t = T_{av} = \frac{1}{\lambda}$$

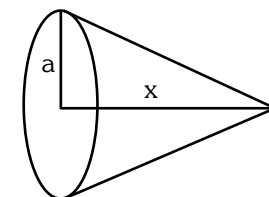
$$So = 1 - e^{-\lambda/\lambda} = 1 - e^{-1} = \frac{e-1}{e}$$

$$37. \therefore C_{eq.} = \frac{C_1 \times C_2}{C_1 + C_2}$$

$$C_{eq.} = \frac{0.9 \times 0.1 \times 10^{-12}}{1 \times 10^{-6}} = 0.09 \mu\text{F}$$



$$38. \therefore B = \frac{\mu_0}{2} \frac{Ia^2}{(a^2 + x^2)^{3/2}}$$



When, $x \gg a$

$$\text{Then, } B = \frac{\mu_0}{2} \frac{Ia^2}{(x^2)^{3/2}} = \frac{\mu_0}{2} \frac{Ia^2}{x^3}$$

Magnetic field on axis of dipole

$$B = \frac{\mu_0}{2\pi} \frac{M}{x^3}$$

$$B \propto \frac{1}{x^3}$$

$\therefore$ Statement I correct and statement II incorrect.

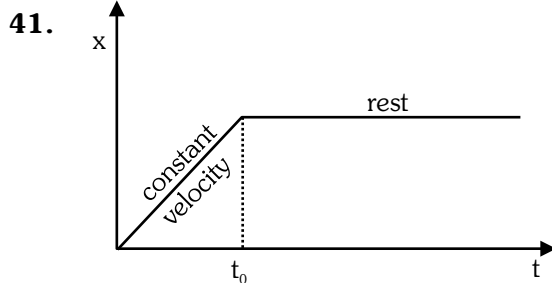
39. When switch S close then in $-\frac{1}{2}$ cycle diode D becomes F.B. and behave as wire. So current doesn't flow from R and output zero.

40. $\frac{1}{2}mv^2 = \frac{kq_1q_2}{r}$

$$\Rightarrow r \propto \frac{1}{v^2}$$

$$\frac{r_1}{r_2} = \left(\frac{v_2}{v_1}\right)^2 = \left(\frac{2v}{v}\right)^2 = 4$$

$$\Rightarrow r_2 = \frac{r_1}{4} = \frac{r}{4}$$



42. $\therefore V_{rms} = \sqrt{\frac{3RT}{M}}$

$$\therefore V_{Ne} = V_{H_2}$$

$$\Rightarrow \sqrt{\frac{3RT}{20.2}} = \sqrt{\frac{3R(288)}{2}}$$

$$\Rightarrow \frac{T}{20.2} = \frac{288}{2}$$

$$\Rightarrow T = 2.9 \times 10^3 \text{ K}$$

43. Work done $= 8\pi T(r_2^2 - r_1^2)$

$$= 8\pi T(4r^2 - r^2) = 24\pi r^2 T$$

44. $\oint \vec{B} \cdot d\vec{s} = 0$

$\therefore$ Magnetic monopoles do not exist.

45. According to question

$$f = 500 \times 10^{12} \text{ Hz}$$

$$d = 0.2 \times 10^{-3} \text{ m}$$

$$D = 1 \text{ m}$$

$$n = 10$$

$$\beta = \frac{n\lambda D}{d} \text{ and } f = \frac{c}{\lambda} \Rightarrow \lambda = \frac{c}{f}$$

$$\text{So, } \beta = n \times \frac{c}{f} \cdot \frac{D}{d}$$

$$= 10 \times \frac{3 \times 10^7}{500 \times 10^{12}} \times \frac{1}{0.2 \times 10^{-4}} = 0.03 \text{ m} = 30 \text{ mm}$$

46. According to question

$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t - \pi/2) = -I_m \cos \omega t$$

$$\therefore P_{inst.} = VI$$

$$\therefore P_{inst.} = -(V_m \sin \omega t)(I_m \cos \omega t) \times \frac{2}{2}$$

$$= \frac{-V_m I_m}{2} \sin(2\omega t)$$

47. $\frac{\ell_1}{\ell_2} = \frac{30}{70} = \frac{X}{Y}$

$$\Rightarrow X = \frac{3}{7} Y \quad \dots(1)$$

$$\frac{50}{50} = \frac{X}{YR/R+Y} = \frac{1}{1}$$

$$\Rightarrow X = \frac{YR}{R+Y} \quad \Rightarrow \frac{3}{7} Y = \frac{YR}{Y+R}$$

$$\Rightarrow 3(Y+R) = 7R \quad \Rightarrow 3Y + 3R = 7R$$

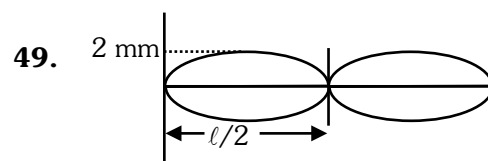
$$\Rightarrow 3Y = 4R$$

$$\Rightarrow Y = \frac{4R}{3} = \frac{4 \times 16}{3} = \frac{64}{3}$$

48. According to question

$$\therefore T^2 \propto R^3$$

$$\therefore \frac{T^2}{x^2} \propto \frac{R^3}{64R^3} \Rightarrow x^2 = 64T^2 \Rightarrow x = 8T$$



$$\ell = \lambda; 2A = \text{amplitude at anti node} = 2\text{mm}$$

$$\text{Amplitude} = 2A \sin kx$$

$$= 2\text{mm} \sin \frac{2\pi}{\lambda} \left(\frac{\ell}{8} \right)$$

$$= 2\text{mm} \sin \left(\frac{\pi \lambda}{\lambda 4} \right) = 2\text{mm} \sin \left(\frac{\pi}{4} \right)$$

$$= \frac{2\text{mm}}{\sqrt{2}} = \sqrt{2} \text{ mm}$$

50. $g = 4\pi^2 \frac{L}{T^2}$

$\therefore \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}$

51.

Polymer	Monomer
Bakelite	Phenol and formaldehyde
Nylon 6, 6	Hexamethylenediamine and adipic acid
Nylon 6	Caprolactam
Terylene	Ethylene glycol and Benzene-1, 4-dicarboxylic acid

52. Ga^+ is more stable than Al^+ as inert pair effect becomes more effective down the group.

53. All three C–O bonds are equal in length with a bond order in between 1 and 2 due to resonance.

54. An electron in Hydrogen atom will have least amount of energy because value of n (principal quantum number) is least which is 1.

55. Volume of the gas is due to the large number of molecules of the gas is not correct because volume of the gas is equal to volume of container.

56. $W = -2.303 nRT \log \left(\frac{p_1}{p_2} \right)$

$W = -2.303 \times 1 \text{ mol} \times 8.314 \frac{\text{J}}{\text{Kmol}} \times 300\text{K} \times \log \left(\frac{5 \text{ atm}}{1 \text{ atm}} \right)$

$W = -2.303 \times 8.314 \times 300 \times 1 \times \log 5$

$W = -2.303 \times 8.314 \times 300 \times 0.6989$

$W = -4014.58 \text{ J}$

57. LiF is sparingly soluble in water because it has high lattice enthalpy which dominates hydration enthalpy.

58. $BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

Top to bottom, molecule weight increases so boiling point increases but due to H-Bonding in NH_3 , the order becomes

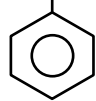
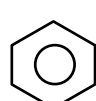
$BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

59. All the domains get oriented in the direction of magnetic field.

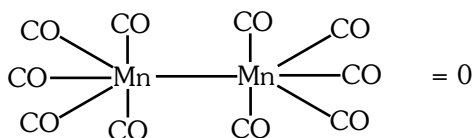
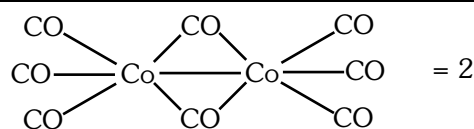
↑ ↑ ↑ ↑ ↑ ↑

60. Because last carbon (asymmetrical) has –OH group on left hand side.

61.

	Amines	pK_b
(a)	$CH_3-\ddot{N}-CH_3$ H	3.27
(b)	$\ddot{N}H_3$	4.75
(c)	$H-\ddot{N}-CH_3$ 	9.30
(d)	 - $\ddot{N}H_2$	9.38

62.

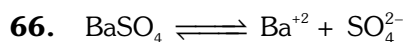
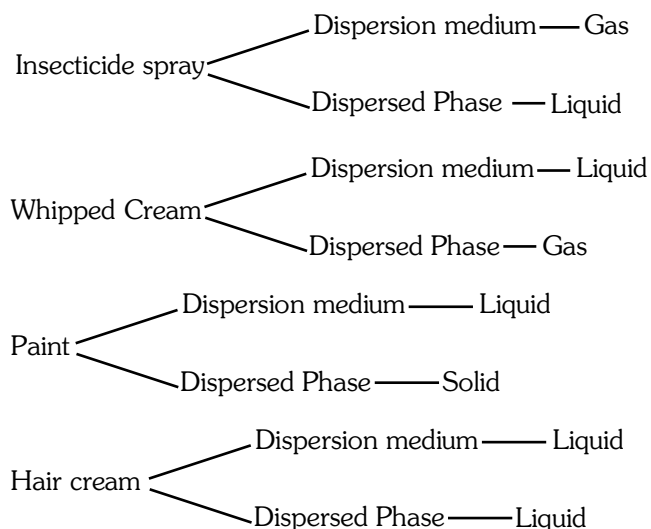


63.

(a)	Element which exhibits +3 oxidation state only	(iii)	Sc $[1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1]$ $Sc^{+3} [1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^0]$
(b)	Element which exhibits more number of oxidation states	(i)	Mn $[1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5]$ (oxidation state from +2 to +7)
(c)	Element which is a reducing agent in its +2 oxidation state	(iv)	Cr $[1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5]$ $Cr^{+2} [1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^4]$ Cr^{+2} acts as a reducing agent, Cr^{+3} is stable because of its configuration - $t_{2g}^3 e_g^0$.
(d)	Element which is not considered as a transition element	(ii)	Zn $[1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}]$ $Zn^{+2} [1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{10}]$ as it is pseudo transition element.

64. Bromination product depends on nature of solvent. CHCl_3 gives monosubstituted bromo derivative as CHCl_3 is a solvent of low polarity whereas reaction of phenol with bromine water yields trisubstituted bromo derivative of phenol.

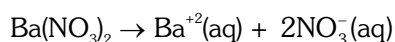
65.



$$K_{sp} = [\text{Ba}^{+2}][\text{SO}_4^{2-}]$$

$$= 1.5 \times 10^{-9} = S^2$$

$$S = \sqrt{1.5 \times 10^{-9}} = 3.87 \times 10^{-5}$$



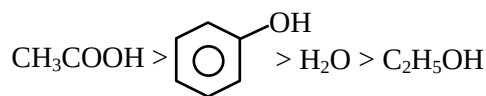
$$[\text{Due to common ion}] \rightarrow (0.1 + S') \quad (S') = K_{sp}$$

$$(0.1) \times S' = 1.5 \times 10^{-9} \quad (\because 0.1 \gg S')$$

$$S' = 1.5 \times 10^{-8} \text{ M}$$

67. (a) Alcohols are weaker acids than water because alkoxide are better proton acceptor than hydroxide.
- (b) Phenols are better acids than alcohols because phenoxide ion is more stable by resonance.

- (c) CH_3COOH is acid, with acid strength greater than all and have $\text{pK}_a = 4.76$



68. $\text{N}_2 = (\text{B.O.} = 3)$

$$\text{O}_2 = (\text{B.O.} = 2)$$

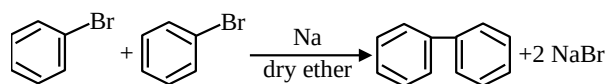
$$\text{O}_2^- = (\text{B.O.} = 1.5)$$

$$\text{O}_2^{2-} = (\text{B.O.} = 1)$$

$$\text{Bond order} \propto \text{Bond enthalpies}$$

$$\text{So, } \text{N}_2 > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$$

69.



Fittig reaction

70. Aldehydes and ketones having at least one α -hydrogen undergo a reaction in the presence of dilute alkali as catalyst to form β -hydroxy aldehydes (aldol) or β -hydroxy ketones (ketol), respectively. This is known as Aldol reaction.
71. (1) It is a true statement
(2) It is a true statement
(3) It is incorrect or false statement.
(4) It is a also a true statement
72. Relative lowering in vapour pressure = Mole fraction of solute

$$\begin{aligned} \therefore x_{\text{solute}} &= \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}} \\ &= \frac{1}{1+3} \\ &= \frac{1}{4} = 0.25 \end{aligned}$$

73. List-I

Elements

(a) O

(b) C

(c) B

(d) N

$B > C > N > O$

As we move across the period from left to right atomic radius decreases due to increase in nuclear charge.

74. Boiling point increases with increase in number of carbon atoms and decreases with branching.

Option (a) has 5 carbon atoms but

Option (b) has 6 carbon atoms

∴ B.P. of (b) > (a)

Option (c) and (d) have '5' carbon atoms each but option (c) has one branch and option (d) has two branches.

∴ B.P. of (c) > (d)

∴ Order of B.P. : $b > a > c > d$

75. The alkali metals and their salts impart characteristic colour in an oxidising flame. This is because the heat from the flame excites the valence electron which comes back to the ground state to emit light of wavelength in visible region.

76. The species which can undergo both oxidation and reduction exhibit disproportionation reactions.

i.e., If the central atom is the species which can show both increase and decrease in O.N. then it shows disproportionation reaction.

Clearly Option 2, 3, 4 are not possible as they all have ClO_4^- as common species which is having Cl already in +7 oxidation state.

77. Rate = $K[A]^2$

List-II

Atomic radii (pm)

(iii) 66 pm

(iv) 77 pm

(i) 88 pm

(ii) 74 pm

(d) If the concentration of reactant is doubled the rate of reaction will increase four times as per the rate equation.

$$\text{i.e., rate} = K[2K]^2$$

$$\text{rate} = 4K$$

(b) Rate constant is independent of concentration but temperature dependent.

78. Knowledge based

79. (a) 4.48 litres of O_2 at STP $\rightarrow$ 6.4 g

$$n_{\text{O}_2} = \frac{V}{22.4} = \frac{4.48}{22.4} = 0.2 \text{ moles}$$

$$w = n \times M_{\text{O}_2} = 0.2 \times 32 = 6.4 \text{ g}$$

(b) 12.022×10^{22} molecules of $\text{H}_2\text{O} \rightarrow$ 0.2 moles

$$n_{\text{H}_2\text{O}} = \frac{N}{N_A} = \frac{12.022 \times 10^{22}}{6.022 \times 10^{23}} = 2 \times 10^{-1} = 0.2 \text{ moles}$$

(c) 96 g of $\text{O}_2 \rightarrow$ 67.2 litres at STP

$$\frac{w}{M} = \frac{V}{22.4}$$

$$\frac{96}{32} = \frac{V}{22.4} \quad V = \frac{96 \times 22.4}{32} = 67.2 \text{ litres}$$

(d) 88 g of $\text{CO}_2 \rightarrow 12.044 \times 10^{23}$ molecules

$$\frac{w}{M} = \frac{N}{N_A} \quad \therefore \frac{88}{44} = \frac{N}{6.022 \times 10^{23}}$$

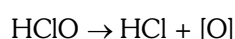
$$N = 6.022 \times 10^{23} \times 2$$

$$= 12.044 \times 10^{23} \text{ molecules}$$

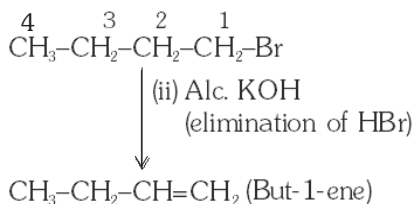
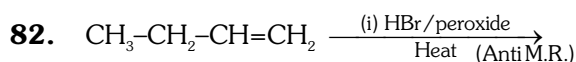
80. Knowledge based

81. Chlorine reacts with water to form HCl and HClO.

But HClO (hypochlorous acid) is unstable and easily dissociates to form nascent oxygen



Nascent oxygen is a more powerful oxidising agent which is responsible for the bleaching nature.



83. formula $\Delta G^\circ = -nFE^\circ_{\text{cell}}$

(a) $n = 6 \quad \therefore \Delta G^\circ = -6 \times F \times 0.404$
 $= -2.424 F$

(b) $n = 2 \quad \therefore \Delta G^\circ = -2 \times F \times 1.211$
 $= -2.422 F$

(c) $n = 3 \quad \therefore \Delta G^\circ = -3 \times F \times 0.807$
 $= -2.421 F$

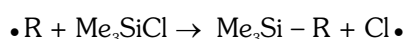
84. Order of reactivity of alcohols towards acid catalyst dehydration is

$$3^\circ > 2^\circ > 1^\circ$$

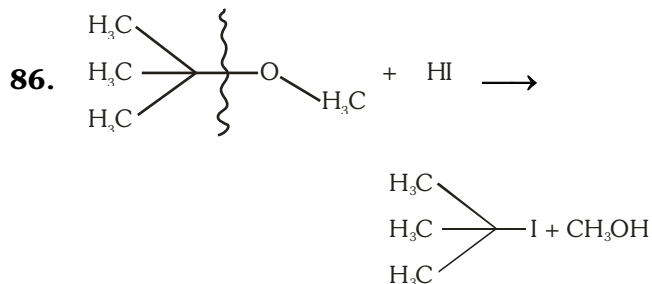
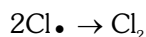
$\therefore a < c < d$ but option (2) is least because OH group is attached sp^2 hybridised carbon atom and partial bond character develops between C-OH bond.

85. The chain length of the silicone polymer can be controlled by adding Me_3SiCl [CH_3]₃SiCl.

The reaction of the sequence can be represented as :

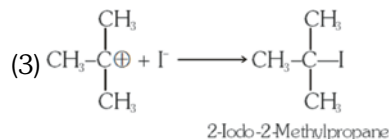
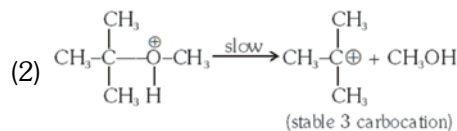
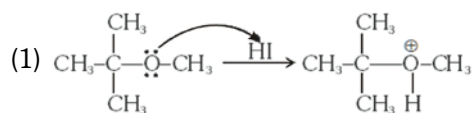


This blocks the end of the silicon polymer and thus terminates the chain propagation.



Whenever one of the alkyl group is a tertiary group, the halide formed is a tertiary halide.

Mechanism :



87. The amount of BOD in water is a measure of the amount of organic matter in the water, in terms of how much oxygen will be required to break it down biologically.

If BOD < 5 ppm (clean water)

If BOD ≥ 17 ppm (highly polluted water)



$t=0, n = \quad 1 \quad 1 \quad 0 \quad 0$

$t=t_{\text{eq}}, n = \quad 1-x \quad 1-x \quad x \quad x$

Moles of C at equilibrium = $x = 0.7$

$\therefore$ Concentrations at equilibrium are :

$$[\text{A}] = \frac{n}{V} = \frac{1-x}{1} = 1 - 0.7 = 0.3$$

$$[\text{B}] = \frac{n}{V} = \frac{1-x}{1} = 1 - 0.7 = 0.3$$

$$[\text{C}] = \frac{n}{V} = \frac{x}{1} = 0.7$$

$$[\text{D}] = \frac{n}{V} = \frac{x}{1} = 0.7$$

$$K_C = \frac{[\text{C}][\text{D}]}{[\text{A}][\text{B}]} = \frac{0.7 \times 0.7}{0.3 \times 0.3} = \frac{49}{9} = 5.4 \text{ (approx)}$$

89. We know, $\Lambda_m = \frac{\kappa \times 1000}{M}$

On putting values,

$$\Lambda_m = \frac{(4.95 \times 10^{-5}) \times 1000}{0.001028}$$

$$= 48.15 \text{ S cm}^2 \text{ mol}^{-1}$$

Now, $\Lambda_m^\circ = \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl})$

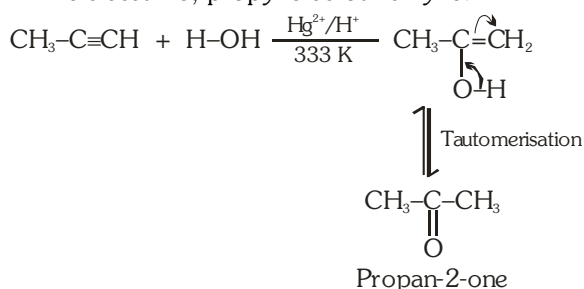
$$= 91.05 + 425.9 - 126.4$$

$$= 390.55 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{48.15}{390.55} = 0.123$$

90. Hydration of Ethyne gives aldehyde but hydration of alkynes having more than 2 carbons gives ketone.

Lets assume, propyne as our alkyne.

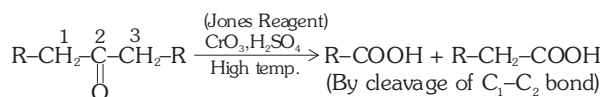


91. Ellingham diagram provides an idea or gives a prediction about the feasibility of thermal reduction of an ore.

Ellingham diagram do not deal with the kinetics of a reduction reaction.

92. Ketones are oxidised under vigorous conditions i.e., strong oxidising agents and at elevated temperature.

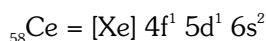
Oxidation involves carbon-carbon cleavage to form mixture of carboxylic acids having less number of carbon atoms than parent ketone



Cleavage of C₂-C₃ bond also gives same product.

93. ${}_{63}\text{Eu} = [\text{Xe}]4f^7 6s^2$

Eu^{2+} is a strong reducing agent changing to common +3 state.



Ce(IV) is a strong oxidant.

$$E^\circ \text{ of } \text{Ce}^{4+}/\text{Ce}^{3+} = 1.74 \text{ V}$$

The E_r° of Ce suggests that it can oxidise water.

94. (P) Power = 1 mW = 10^{-3} watt

$$E = P \times t = 10^{-3} \times 0.1 = 10^{-4} \text{ J}$$

$$\lambda = 1000 \text{ nm} = 1000 \times 10^{-9} \text{ m} = 10^{-6} \text{ m}$$

We know,

$$E = nh\nu$$

Here, E = Energy of n photons

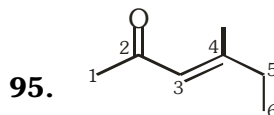
n = number of photons

$h\nu$ = energy of 1 photon

$$10^{-4} = n \times \frac{hc}{\lambda}$$

$$10^{-4} = \frac{n \times (6.626 \times 10^{-34}) \times (3 \times 10^8)}{10^{-6}}$$

$$n = 5 \times 10^{14} \text{ photons}$$



Longest parent chain is selected by giving lowest numbering to ketone.

IUPAC name – 4-methyl hex-3-en-2-one

96. Moles of gas is conserved

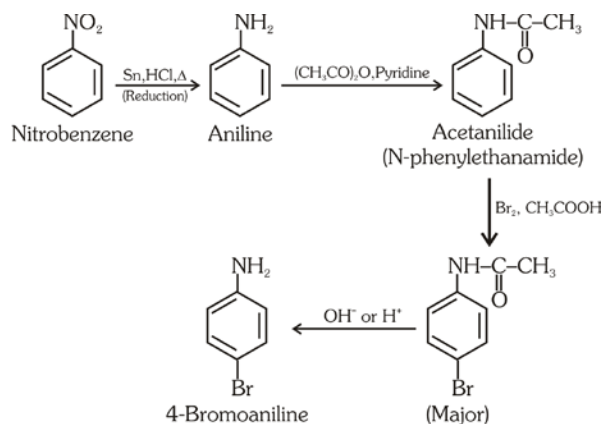
$$n_{\text{total}} = n_A + n_B$$

$$\text{Assuming gases as ideal, } n = \frac{PV}{RT}$$

$$\frac{PV}{RT} = \frac{P_1V_1}{RT} + \frac{P_2V_2}{RT}$$

$$P_{\text{total}} = \frac{P_1V_1 + P_2V_2}{V_{\text{total}}} = \frac{1 \times 250 + 2 \times 500}{1000 \text{ ml}} = 1.25 \text{ bar}$$

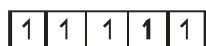
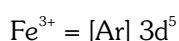
- 97.



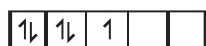
98. $\text{Cl}^- \rightarrow$ Weak field ligand (no pairing occurs)

$\text{H}_2\text{O} \rightarrow$ Weak field ligand (no pairing occurs)

$\text{CN}^- \rightarrow$ Strong field ligand (no pairing occurs)

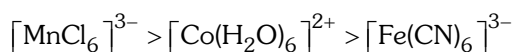


On pairing



no. of unpaired $e^- = 1$

So, spin magnetic moment answer :



99. In case of semiconductor, the gap between valence band and conduction band is small.

Therefore option (4) is incorrect as here it is mentioned conduction band and valence band overlaps which happens in conductors actually.

100. We know, Arrhenius equation

$$\ln K = \ln A - \frac{E_a}{RT}$$

$$y = C + mx$$

$$\text{So, slope} = -\frac{E_a}{R} = -1 \times 10^4 \text{ K}$$

$$\text{So, } E_a = 10^4 R$$

$$= 10^4 \times 8.3$$

$$= 8.3 \times 10^4 \text{ J mol}^{-1}$$